PURBANCHAL UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING
KHWOPA ENGINEERING COLLEGE
LIBALI-08, BHAKTAPUR

MID DEFENCE REPORT ON
Maternal Health Risk Screening Using Random Forest and SHAP

Project work submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering (Seventh Semester)

Submitted by

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With Regards,

Chiryau Shrestha
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Abstract

Maternal mortality remains a pressing public health challenge in Nepal, where the maternal mortality ratio (MMR) stands at 151 per 100,000 live births according to the 2021 National Population and Housing Census, with significant disparities observed across provinces such as Lumbini (207) and Karnali (172) due to geographic isolation, inadequate healthcare infrastructure, and socioeconomic barriers. In this project, we propose the design and development of a web-based, offline-first Artificial Intelligence (AI) screening and automated triage platform intended to assist Female Community Health Volunteers (FCHVs) in identifying and managing maternal health risks in rural Nepal. The proposed system leverages a client-side machine learning model, trained on the UCI Maternal Health Risk Dataset and CDC Natality, which classifies pregnant women into low, mid, and high risk categories based on six clinical parameters, namely age, systolic blood pressure, diastolic blood pressure, blood sugar, body temperature, and heart rate, using a Random Forest classifier achieving high accuracy. To address the challenge of limited internet connectivity in remote areas, the application is designed as a Progressive Web App (PWA) using TensorFlow.js for client-side inference and PouchDB for offline-first local storage, which synchronizes with a FastAPI cloud backend when connectivity is restored. Upon detection of a high-risk pregnancy, the system automatically triggers an SMS alert through the Sparrow SMS API to notify the nearest Comprehensive Emergency Obstetric and Neonatal Care (CEONC) facility and the patient's family, thereby addressing the critical three delays in maternal care.This project aims to contribute a scalable, community-driven digital health solution that bridges the gap between rural populations and clinical care, and supports Nepal's national maternal health improvement goals.

Keywords: Maternal Health, Artificial Intelligence, Risk Screening, Progressive Web App, Offline-First, Female Community Health Volunteers, Nepal, TensorFlow.js, PouchDB, Sparrow SMS

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Chapter 1

Introduction

1.1 Background

Maternal health remains a critical public health concern, particularly in developing countries like Nepal, where access to timely and quality healthcare services is limited in rural and remote areas. Despite improvements in healthcare infrastructure and government initiatives, maternal mortality and complications during pregnancy continue to pose significant challenges. Many pregnant women in rural regions rely on Female Community Health Volunteers (FCHVs), who play a vital role in providing basic healthcare services, monitoring maternal conditions, and facilitating referrals to health institutions.

However, FCHVs often lack advanced tools and decision-support systems to accurately assess maternal health risks during routine visits. Their evaluations are primarily based on manual observations and basic guidelines, which may lead to delayed identification of high-risk cases. Early detection of complications such as hypertension, gestational diabetes, and abnormal vital signs is crucial for reducing maternal and neonatal mortality.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is now possible to develop intelligent systems that can assist healthcare workers in making data-driven decisions. By analyzing key health indicators, AI models can predict risk levels and provide actionable insights, thereby improving the efficiency and accuracy of maternal health assessments. This project aims to leverage AI technologies to support FCHVs in rural Nepal by providing an accessible and reliable maternal health risk screening and referral support system.

1.2 Statement of problems

In rural areas of Nepal, maternal health assessment is often limited by the lack of diagnostic tools, insufficient training, and absence of real-time decision support systems for health workers. Female Community Health Volunteers (FCHVs), who serve as the primary point of contact for pregnant women, rely on manual methods and subjective judgment to identify potential risks. This approach can lead to inaccurate or delayed detection of high-risk pregnancies.

Additionally, there is no widely available, Nepal-specific digital system that integrates patient data, performs automated risk analysis, and provides immediate recommendations for referral or monitoring. The absence of such systems results in missed opportunities for early intervention, ultimately contributing to preventable maternal and neonatal complications.

Therefore, there is a need for an intelligent, easy-to-use, and accessible system that can assist FCHVs in accurately assessing maternal health risks and guiding timely referral decisions, even in low-resource settings.

1.3 Objectives

To develop an AI-assisted maternal health risk screening and referral support system that helps rural healthcare workers identify high-risk pregnancies and take appropriate actions.

1.4 Significance and scopes

1.4.1 Significances

- Helps in early detection of high-risk pregnancies, enabling timely medical intervention and reducing maternal and neonatal complications in rural areas.
- Provides Female Community Health Volunteers with an AI-based decision support tool, improving accuracy in maternal risk assessment.
- Demonstrates the practical application of Machine Learning and Explainable AI (SHAP) in healthcare, contributing to innovation in AI-driven medical systems.

1.4.2 Scopes

- Focuses on building a machine learning model to predict maternal risk levels using health parameters such as blood pressure, blood sugar, and heart rate.
- Includes a web-based interface that provides risk predictions, explainability (SHAP), and advisory outputs in Nepali and English.
- Limited to publicly available datasets and a prototype system, with future scope for Nepal-specific data integration and mobile deployment.

Chapter 2

Literature Review

2.1 Maternal Mortality and Health Challenges in Nepal

Maternal mortality remains a critical public health concern in Nepal, with the maternal mortality ratio (MMR) recorded at 151 per 100,000 live births in the 2021 National Population and Housing Census [1]. Significant provincial disparities persist, with Lumbini Province recording the highest MMR at 207 and Bagmati Province the lowest at 98, reflecting the unequal distribution of healthcare infrastructure across geographic regions [2, 3]. The postpartum period accounts for approximately 61

2.2 Community Health Volunteers and Digital Health Interventions

Female Community Health Volunteers (FCHVs) serve as the primary health contact for millions of rural women in Nepal who are unable to access formal health facilities due to distance, cost, or cultural barriers. Research has consistently highlighted the FCHV program as one of Nepal's most impactful community health mechanisms, with FCHVs occupying a trusted position that makes them effective agents for maternal screening and referral [4, 5]. However, the absence of standardized digital triage tools limits their ability to identify and respond to high-risk pregnancies in a timely manner. Studies on digital health interventions in low- and middle-income countries (LMICs) have demonstrated that equipping community health workers with AI-assisted mobile tools can improve risk identification and referral processes, provided that the tools are designed for resource-constrained environments and are accessible to users with varying levels of digital literacy [6, 7].

2.3 Machine Learning for Maternal Risk Classification

Several studies have evaluated machine learning classifiers for maternal health risk prediction using structured clinical datasets. Research utilizing the UCI Maternal Health Risk Dataset, containing 1,014 clinical samples with features including age, blood pressure, blood sugar, body temperature, and heart rate, has shown that Random Forest classifiers achieve approximately 88

2.4 Offline-First Architecture and Automated Alerting

Progressive Web Apps (PWAs) have emerged as an appropriate architecture for health applications in low-connectivity rural environments, enabling application functionality without continuous internet access through mechanisms such as Service Workers and local storage. PouchDB, an in-browser database built on IndexedDB, can provide offline data persistence and synchronization with cloud backends when network connectivity is restored. TensorFlow.js enables machine learning model inference to run within the browser, reducing dependence on server-side API calls during risk assessment. These approaches are particularly relevant to rural and resource-limited healthcare environments where continuous connectivity cannot be guaranteed [6, 7].

In parallel, SMS-based automated alerting can help address referral delays in obstetric emergencies by notifying receiving facilities before patient arrival. In Nepal, Sparrow SMS provides an API-based gateway for programmatic SMS dispatch and has been explored for application integration within the Nepali context [8]. Such an approach can support timely communication between community-level health workers and referral facilities during high-risk maternal cases.

2.5 Research Gap and Justification

While AI-based maternal health tools have been deployed successfully in Kenya, India, and Guatemala, no published implementation has combined client-side TensorFlow.js inference, offline-first PouchDB storage, and automated Sparrow SMS referral alerting within a single multilingual web platform tailored specifically for Nepal's FCHV workforce. Existing models have largely been trained on non-Nepali datasets without localized validation, and current digital health deployments in Nepal remain cloud-dependent and may be inaccessible in areas with poor network coverage [6, 7]. The importance of adapting AI-based maternal risk prediction to low-resource environments and improving interpretability has also been highlighted in recent research [9].

This project directly addresses these gaps by proposing an integrated, offline-capable AI screening platform aligned with Nepal's maternal health infrastructure and community health workforce. The proposed system combines machine learning-based maternal risk classification with explainable predictions, offline data handling, multilingual accessibility, and automated referral communication, thereby providing a practical framework for supporting FCHVs in early identification and management of high-risk pregnancies.

Chapter 3

Project Management

3.1 Team Members

The project will be carried out with the contribution of the following four team members.

1. Chirayu Shrestha (790311)
2. Janam Gaiju (790314)
3. Krijal Gwachha (790315)
4. Nikesh Manandar (790320)

3.2 Work Breakdown Structure

S.N	Job Description	Duration (days)	Week			
			1^{st} - 3^{rd}	4^{th} - 6^{th}	7^{th} - 9^{th}	10^{th} - 12^{th}
1.	Problem Identification	6				
2.	Analysis	11				
3.	Design	14				
4.	Coding	28				
5.	Documentation	30				

3.3 Software Requirements

- **Programming Languages:** Python, JavaScript
- **Frameworks and Libraries:** FastAPI, React.js, Scikit-learn, XGBoost, TensorFlow (optional), SHAP ,Django
- **Database:** Pouch DB, PostgresSQL
- **Development Tools:** VS Code
- **APIs:** LLM API (for advisory generation)
- **Operating System:** Windows
- **Browser:** Google Chrome or any modern web browser

3.4 Hardware Requirements

- **Processor:** Minimum Intel i5 or equivalent
- **RAM:** Minimum 8 GB (16 GB recommended)
- **Storage:** At least 10–20 GB free space
- **Devices:** Laptop/Desktop for development, optional mobile device for testing
- **Internet:** Required for API usage and setup (optional for offline features)

3.5 Functional Requirements

- **Authentication:** User authentication system (PIN/login for health workers)
- **Data Input:** Module to enter patient health data (age, blood pressure, sugar level, etc.)
- **Prediction:** Risk prediction using trained machine learning model
- **Output Display:** Display of risk level (Low / Medium / High) with confidence score
- **Explainability:** SHAP-based explanation showing key contributing factors
- **Advisory:** Advisory generation in Nepali and English
- **Referral Support:** Referral suggestion for high-risk cases (nearest hospital guidance)
- **Data Storage:** Patient data storage and tracking across multiple visits
- **Visualization:** Visualization of patient risk trends over time

3.6 Non-Functional Requirements

- **Usability:** Simple and user-friendly interface for FCHVs
- **Performance:** Fast response time (within a few seconds)
- **Reliability:** Consistent and accurate system behavior
- **Scalability:** Ability to handle increasing users and data
- **Security:** Secure storage of patient data and authentication
- **Availability:** Accessible when needed, with potential offline support
- **Maintainability:** Easy to update models and system components

Chapter 4

Feasibility Study

4.1 Technical Feasibility

The proposed system is technically feasible as it uses widely available technologies such as Python, FastAPI, and React.js. Machine learning models like Random Forest and XGBoost can be implemented using standard libraries such as Scikit-learn. The system does not require high-end infrastructure and can run on standard computers. Integration of SHAP for explainability and Gemini API for advisory generation is also achievable with existing tools and APIs.

4.2 Economic Feasibility

The project is cost-effective as it primarily uses open-source technologies and tools. Development can be carried out using freely available software such as VS Code and open-source libraries. Minimal cost may be involved in API usage and deployment, but overall, the system is affordable and suitable for low-resource settings.

4.3 Operational Feasibility

The system is designed to be user-friendly and suitable for Female Community Health Volunteers (FCHVs) with minimal technical knowledge. The interface will be simple, and the outputs will be easy to understand in both Nepali and English, ensuring effective real-world usability.

4.4 Schedule Feasibility

The project can be completed within the given academic timeline. Development is divided into phases such as data preprocessing, model training, backend and frontend development, and testing. With proper planning and team coordination, the system can be successfully implemented within the semester duration.

Chapter 5

Methodology

5.1 Block Diagram

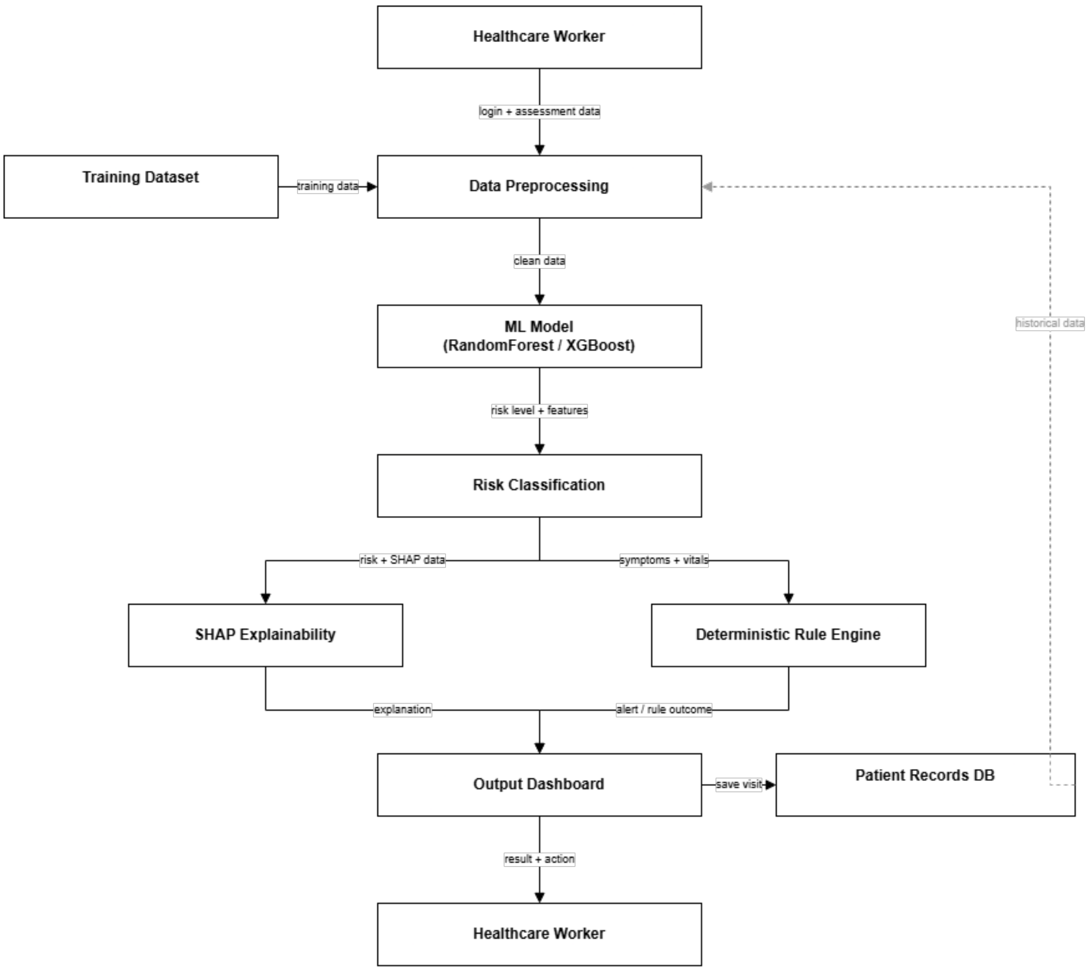


Figure 5.1: Block Diagram of the Proposed System

1. The **Healthcare Worker** logs in and submits a patient's assessment data (vitals and symptoms).

2. **Data Preprocessing** validates and prepares that input. It also pulls in the Training Dataset (used once, offline, to build the model) and receives historical data fed back from the database through the dashed line, so trend context is available.

3. The **ML Model (Random Forest / XGBoost)** runs on the cleaned data and outputs a risk level with feature values.

4. **Risk Classification** takes that output and splits it into two parallel branches:
   - **Left branch: SHAP Explainability** explains why the model produced that risk level.
   - **Right branch: Deterministic Rule Engine** independently checks the same vitals and symptoms against fixed clinical thresholds. This safety layer is separate from the ML model and cannot be overridden by it.
5. Both branches feed into the **Output Dashboard**, where the SHAP explanation and the rule engine's alert or outcome are combined into one result.
6. The dashboard saves the visit to the **Patient Records DB**.
7. The same database feeds historical data back to Data Preprocessing in step 2 for future trend calculations.
8. Finally, the dashboard sends the result and action back to the **Healthcare Worker**, closing the loop.

5.2 System Architecture

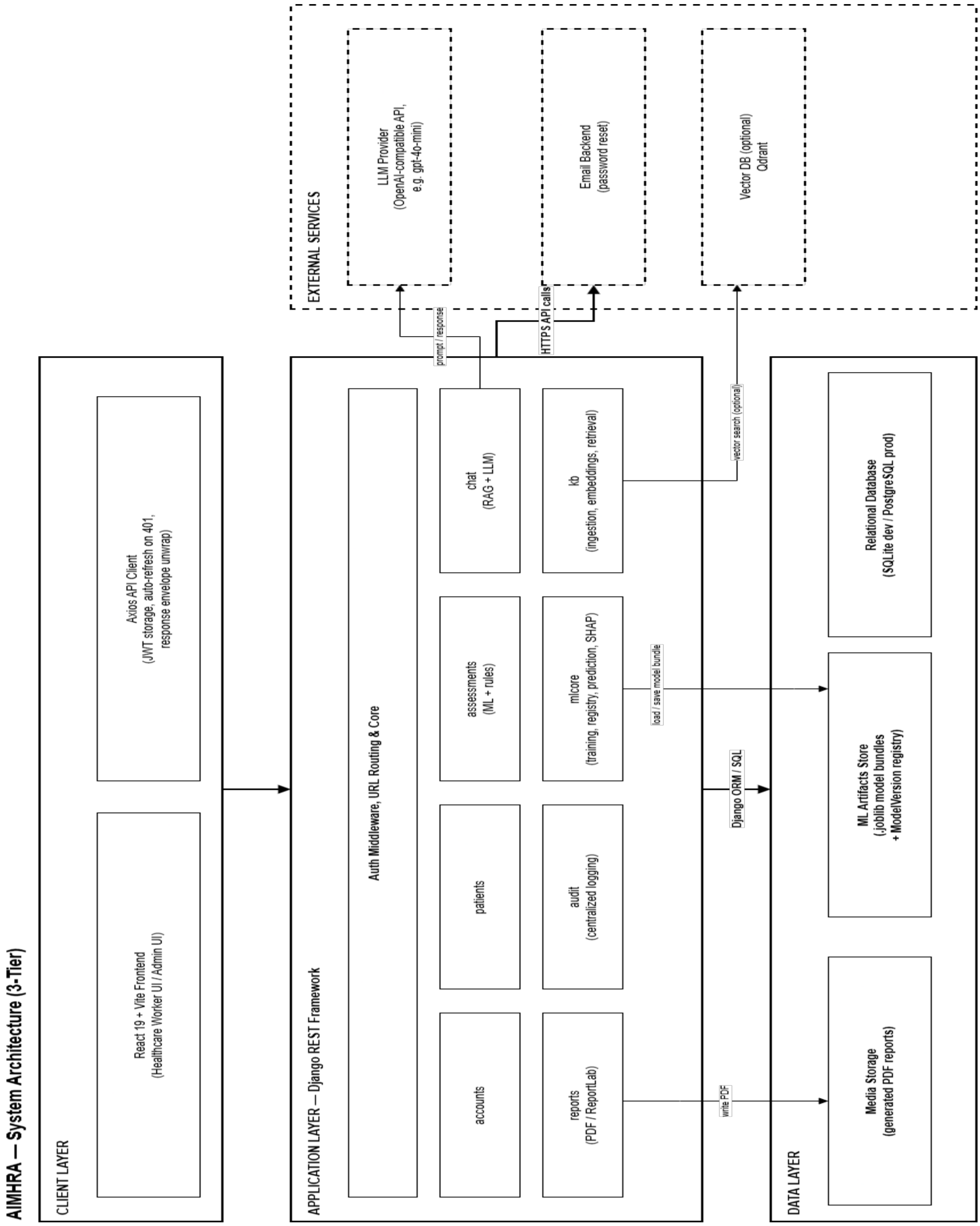


Figure 5.2: System Architecture of the Proposed System

AIMHRA is a 3-tier system:

Client Layer — A React 19 + Vite frontend serves the Healthcare Worker and Admin interfaces. Axios handles all API calls, attaching JWT tokens and auto-refreshing them when they expire.

Application Layer — A Django REST Framework backend does all the real work. A shared core handles authentication, permissions, and response formatting, and then eight independent apps each own one concern: `accounts` (users), `patients` (profiles), `assessments` (the ML + rules risk pipeline), `chat` (RAG + LLM chatbot), `reports` (PDF generation), `audit` (logging), `mlcore` (model training/registry/SHAP), and `kb` (knowledge base ingestion and retrieval). The frontend talks to this layer over HTTPS using REST calls secured with JWT.

Data Layer — A relational database (SQLite for dev, PostgreSQL for production) holds all structured data. Trained ML model files live separately in an artifacts store with their own version registry, and generated PDF reports are kept in media storage. The application layer reaches this through Django's ORM.

External Services — Kept in its own boundary because these are outside the system: an OpenAI-compatible LLM provider (for chatbot answers), an optional Qdrant vector database (for RAG at scale), and an email backend (for password resets). Only the `chat` app talks to the LLM, and only `kb` talks to the optional vector store — everything else stays inside the application/data boundary.

5.3 Use Case Diagram

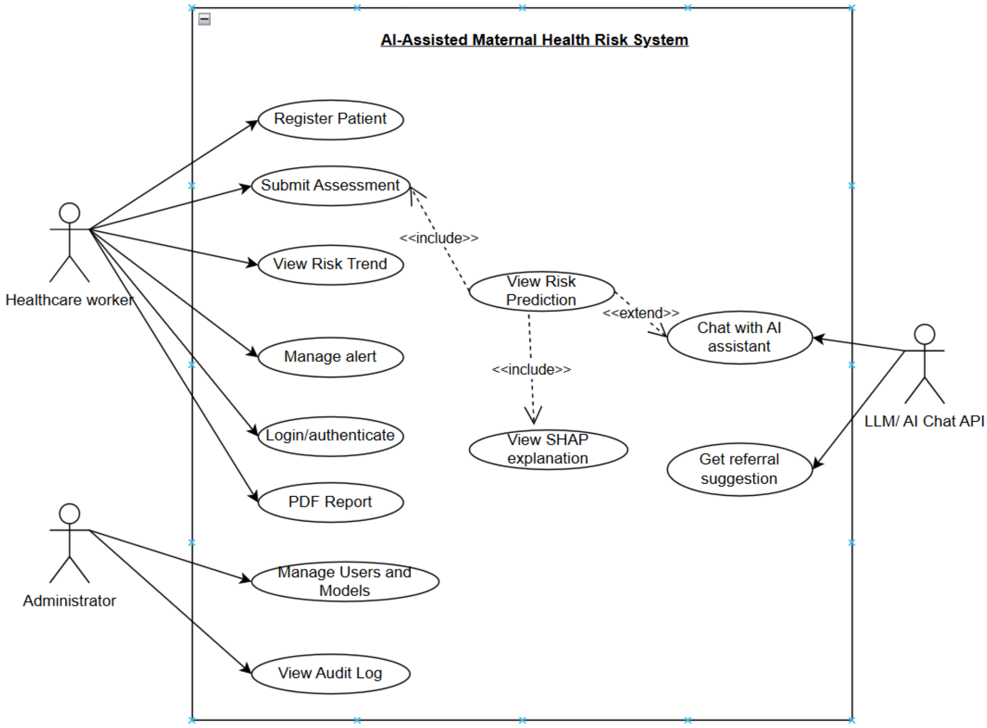


Figure 5.3: Use Case Diagram of the Proposed System

The Use Case Diagram illustrates the interactions between three actors — the Healthcare Worker, the Administrator, and the external LLM/AI Chat API — and the core functionalities of the AI-Assisted Maternal Health Risk System. The Healthcare Worker is the primary actor, with access to patient-facing operations including registering a patient, submitting an assessment, viewing risk trends, managing alerts, logging in, and generating PDF reports. Submitting an assessment includes the View Risk Prediction use case, which in turn includes View SHAP Explanation for model interpretability and extends to Chat with AI Assistant for further guidance, with both the AI chat and the Get Referral Suggestion use case supported by the external LLM/AI Chat API. The Administrator operates independently, with use cases limited to managing users and models and viewing the audit log, reflecting the system's restriction of administrative access to configuration and oversight rather than direct patient data.

5.4 Data Flow Diagram

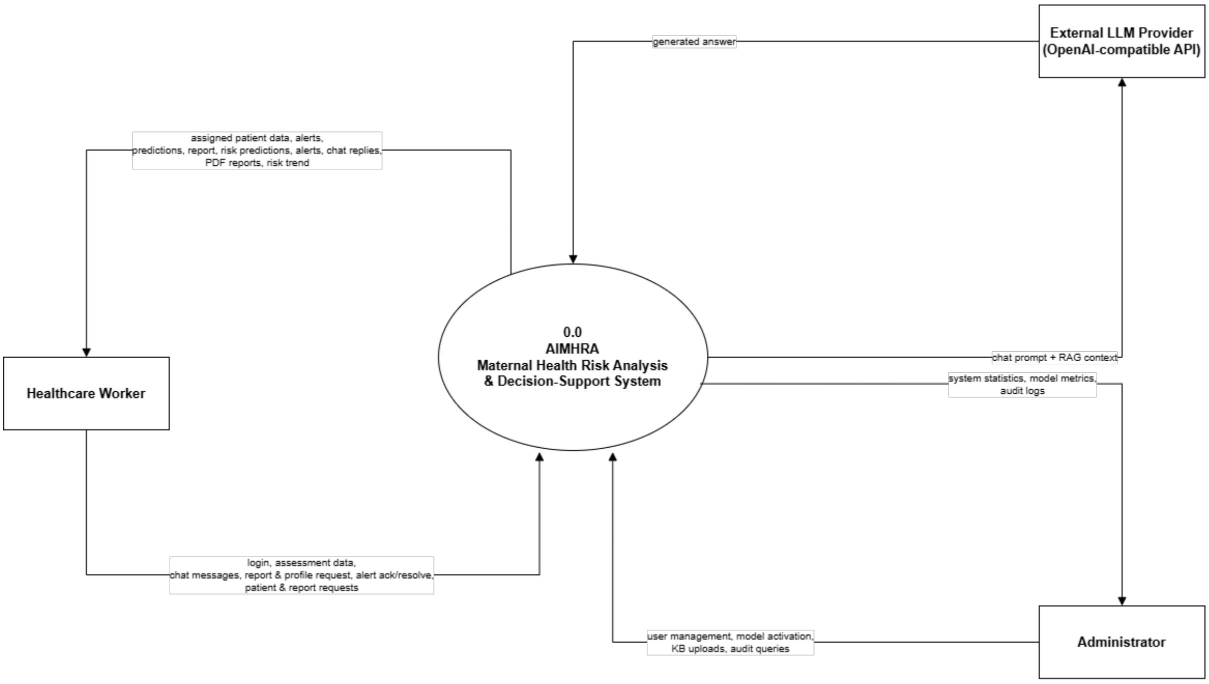


Figure 5.4: Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram presents the system's interactions with three external entities:

- **Healthcare Worker** — the only human user who operates the system day-to-day now that the Patient role is removed. **Sends:** login credentials, new/edited patient data, assessment vitals & symptoms, chat questions, report requests, and alert acknowledgements. **Receives back:** auth tokens, ML risk predictions with SHAP explanations, rule-based alerts, chatbot answers with sources, generated PDF reports, and risk history/trend data.
- **Admin** — manages the system itself rather than individual patients. **Sends:** account creation/deactivation for workers and admins, patient-to-worker assignments, model training/activation commands, and knowledge-base upload/reindex requests. **Receives back:** system statistics, model metrics and comparisons, KB indexing status, and audit log records.
- **LLM Provider (external)** — the only non-human external system, since AIMHRA calls out to an OpenAI-compatible API for chatbot generation. The system sends it a prompt with RAG context and gets a generated answer back.

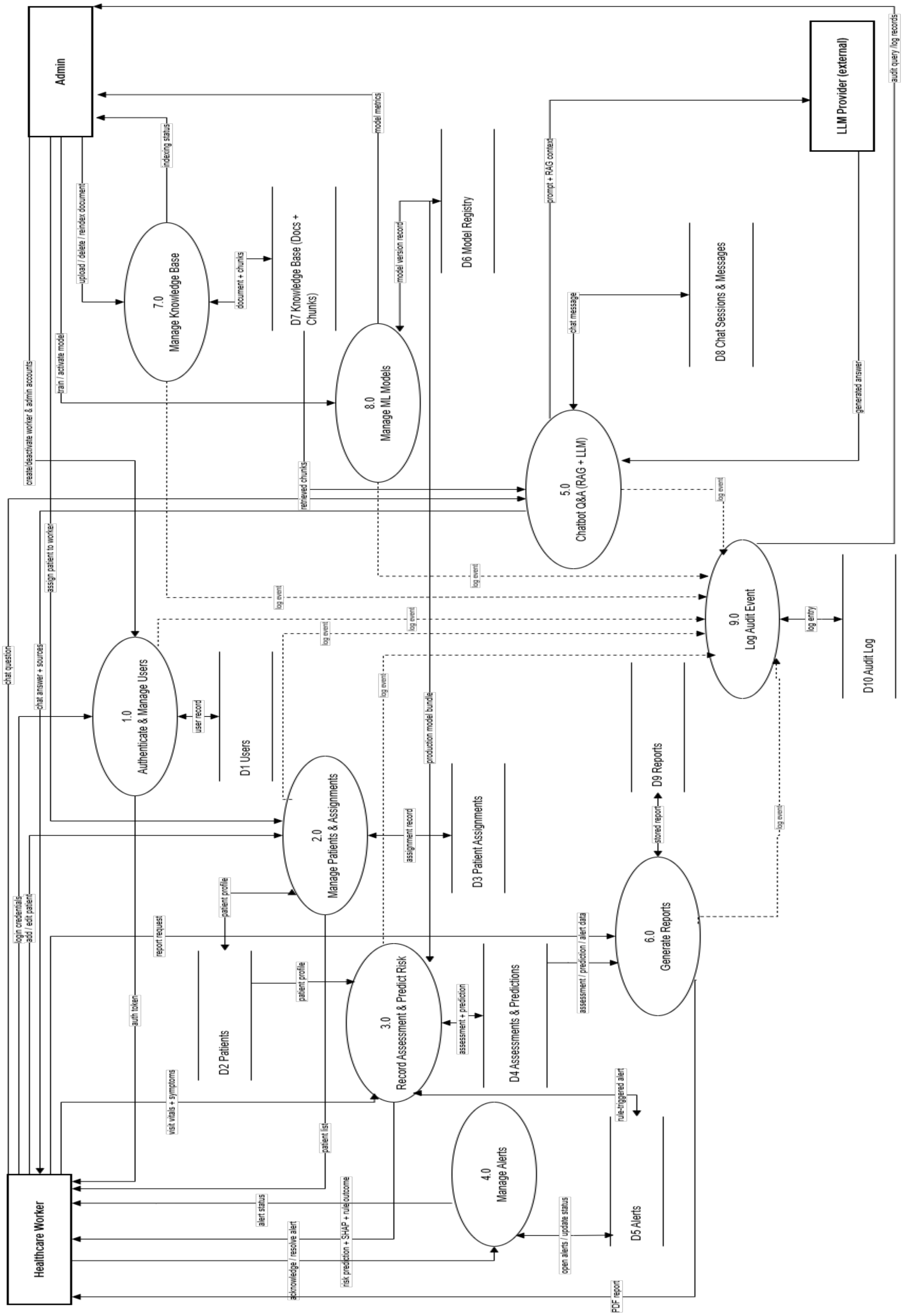


Figure 5.5: Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram expands AIMHRA into the following nine processes:

1. **Authenticate & Manage Users** — handles login for the Healthcare Worker and account creation/deactivation by Admin. Reads/writes D1 Users.

2. **Manage Patients & Assignments** — this is where the role change actually lives: it now takes "add/edit patient" directly from the Healthcare Worker (not just from Admin), writing to D2 Patients, while Admin still separately assigns patients to workers via D3 Patient Assignments.

3. **Record Assessment & Predict Risk** — the core pipeline: takes vitals/symptoms from the Healthcare Worker, pulls the patient's profile from D2, loads the production model from D6 Model Registry, stores the result in D4 Assessments & Predictions, and raises anything into D5 Alerts.

4. **Manage Alerts** — lets the Healthcare Worker acknowledge or resolve open alerts in D5.

5. **Chatbot Q&A (RAG + LLM)** — takes a chat question, retrieves matching chunks from D7 Knowledge Base, stores the conversation in D8 Chat Sessions, and calls out to the external LLM Provider for the generated answer.

6. **Generate Reports** — pulls assessment/prediction/alert data from D4 and produces a PDF, stored in D9 Reports.

7. **Manage Knowledge Base** — Admin-only: upload, delete, or reindex documents into D7.

8. **Manage ML Models** — Admin-only: train and activate models, tracked in D6 Model Registry.

9. **Log Audit Event** — a shared logging process that every other process (1, 2, 3, 5, 6, 7, 8) sends a dashed "log event" flow to, all landing in D10 Audit Log. It's drawn as its own process rather than folded into each one because that's how the actual system works — audit logging is a single centralized service every module calls into, not something each module implements separately.

5.5 Data Collection and Dataset Description

The proposed AI-Assisted Maternal Health Risk Screening and Referral Support System is designed to process patient health measurements entered by the FCHV (Health Worker). Accordingly, the datasets used in this project are selected to support machine learning model training, evaluation, and cross-dataset comparison. The system uses the common features shared across datasets rather than forcing full merging, allowing both independent model training and comparative analysis. The following datasets are proposed for use in this system.

5.5.1 Dataset — Maternal Health Risk Dataset

This study uses a single maternal health risk dataset comprising 6,103 records collected through Internet of Things (IoT)-enabled Medical Cyber-Physical Systems (MCPS) across nine healthcare facilities in Bangladesh between February 2021 and January 2023. The dataset includes eight clinical features — Age, Body Temperature, Heart Rate, Systolic Blood Pressure, Diastolic Blood Pressure, BMI, HbA1c, and Fasting Blood Glucose — together with a target variable representing three expert-validated risk categories: Low, Mid, and High Risk. Data collection employed standardized IoT sensors (Raspberry Pi 4 controllers with medical-grade sensor arrays), and the records underwent rigorous quality validation, with clinically implausible samples removed to yield 6,057–6,058 usable records for model development. This dataset is used independently as the sole training, validation, and testing source for the machine learning models developed in this project, without the need for cross-dataset comparison [10].

5.5.2 Feature Alignment Across Datasets

Since the three datasets vary in their feature sets, the system identifies and uses the common attributes available across all datasets for the core prediction pipeline. The shared features used for model training and comparison are: **Age**, **Systolic Blood Pressure**, **Diastolic Blood Pressure**, **Blood Sugar**, and **Heart Rate**. Extended features present only in Dataset 1 (BMI, diabetes indicator, mental health score) are used in supplementary experiments to assess whether additional features improve prediction accuracy.

5.6 Data Preprocessing

Before the collected data can be used by the AI-Assisted Maternal Health Risk Screening System, appropriate preprocessing steps must be applied to ensure that inputs are clean, consistent, and suitable for model training and inference. The preprocessing pipeline covers both the training datasets and real-time patient inputs entered by the health worker through the application. Since three separate datasets are used, preprocessing is applied independently to each dataset before any cross-dataset comparison is performed.

5.6.1 Handling Missing Values

Missing values present in the datasets are handled through median imputation, applied per risk class to preserve distributional properties. This ensures that no record is discarded and that the model receives complete input vectors during both training and prediction. For the CDC Natality subset, records with missing values in the core shared features are dropped if imputation is not feasible due to structural differences in data collection [11].

5.6.2 Feature Scaling and Normalization

5.6.2.1 Min-Max Normalization

All numerical features are normalized to a [0, 1] range using Min-Max normalization prior to model training. This step ensures that no single feature dominates the model due to differences in measurement scale — for example, blood pressure values (60–180 mmHg) versus body temperature values (35–42 °C). The same scaler fitted on each training dataset is applied consistently to real-time inputs during inference [12].

5.6.3 Class Imbalance Handling — SMOTE

Each dataset exhibits some degree of class imbalance, with the High risk class being underrepresented relative to Low and Mid risk classes. To address this, Synthetic Minority Over-sampling Technique (SMOTE) is applied independently to each dataset during training preparation. SMOTE generates synthetic samples for the minority class by interpolating between existing instances, resulting in a more balanced class distribution and improving the model's ability to detect high-risk cases accurately [13].

5.6.4 Feature Engineering and Encoding

No additional derived features are introduced beyond the clinical measurements already present in the datasets. Ordinal encoding is applied to the risk label column across all three datasets, mapping Low, Mid, and High to integer values 0, 1, and 2 respectively, ensuring a consistent target variable format for multi-class classification. Where Dataset 1's extended features (BMI, diabetes indicator, mental health score) are used in supplementary experiments, they are scaled using the same normalization pipeline.

5.7 Evaluation Metrics

To provide meaningful feedback and assess system performance, the evaluation framework covers model classification accuracy, cross-dataset comparison, and the quality of SHAP-based explanations and advisory outputs generated by the system.

5.7.1 System Data Capture and Processing Pipeline

The system processes structured patient measurements entered through the health worker interface. These numerical inputs are passed through the preprocessing pipeline and into the trained machine learning model for risk classification. The model simultaneously produces a predicted risk level and SHAP feature importance values, which are forwarded to the recommendation module and the Gemini API to generate a natural language advisory. The complete pipeline from data entry to result display is captured and used for performance evaluation.

5.7.2 Classification Performance Metrics

The primary evaluation of the ML models uses standard multi-class classification metrics. **Accuracy** measures the proportion of correctly classified records across all three risk classes. **Precision**, **Recall**, and **F1-Score** are computed per class using macro-averaging to account for class imbalance. A confusion matrix is generated for each model and dataset combination to inspect misclassification patterns, particularly between the Mid and High risk classes where correct identification is clinically critical. Multiple models — including Random Forest and XGBoost — are compared across all three datasets to identify the best-performing configuration.

5.7.3 Semantic Explanation Quality (Cosine Similarity)

For evaluating the quality of SHAP-based textual explanations and Gemini-generated advisories, the system measures semantic relevance using **Cosine Similarity**:

$$\text{Similarity}(A,\ B) = \frac{A \cdot B}{\|A\|\ \|B\|} \tag{5.1}$$

Where:

- A represents the generated advisory or explanation embedding vector.
- B represents the ideal reference explanation embedding vector.

A higher similarity score indicates greater semantic alignment between the system-generated advisory and the reference clinical guidance. This metric validates the contextual relevance of advisory text produced for each risk level.

5.7.4 Risk Prediction Confidence Score

The machine learning model outputs a probability distribution across the three risk classes for each prediction. The confidence score is defined as the maximum class probability:

$$\text{CS} = \max\left(P_{\text{low}},\ P_{\text{mid}},\ P_{\text{high}}\right) \tag{5.2}$$

Where:

- CS is the Confidence Score (0–1).

- P_{low}, P_{mid}, P_{high} are the predicted probabilities for each risk class.

A higher confidence score indicates that the model assigns a clear and decisive risk classification to the patient, supporting the health worker in making triage decisions.

5.7.5 Overall System Performance Score (SPS)

The system combines model accuracy and explanation quality into an overall **System Performance Score (SPS)**:

$$\text{SPS} = 0.6 \times \text{Accuracy} + 0.4 \times \text{Similarity} \tag{5.3}$$

Where:

- Accuracy is the overall multi-class classification accuracy of the ML model.
- Similarity is the mean cosine similarity score of generated advisories against reference texts.

This composite score reflects both the predictive reliability of the model and the quality of explanatory content delivered to the health worker.

5.7.6 Referral Match Score

To evaluate the accuracy of the recommendation module, the system computes a **Referral Match Score (RMS)** by comparing system-recommended actions against expert-validated referral targets:

$$\text{RMS} = \frac{\text{Correct Referrals}}{\text{Total High-Risk Cases}} \times 100 \tag{5.4}$$

Where:

- RMS is the Referral Match Score (%).
- Correct Referrals is the number of cases where the recommended action matched the validated target.
- Total High-Risk Cases is the count of all cases classified as High risk.

A higher RMS indicates that the system correctly identifies and recommends the appropriate care pathway — referral, monitoring, or normal checkup — for patients based on their risk level, directly supporting timely maternal care in rural healthcare settings.

Chapter 6

Outcome

6.1 Admin Interface

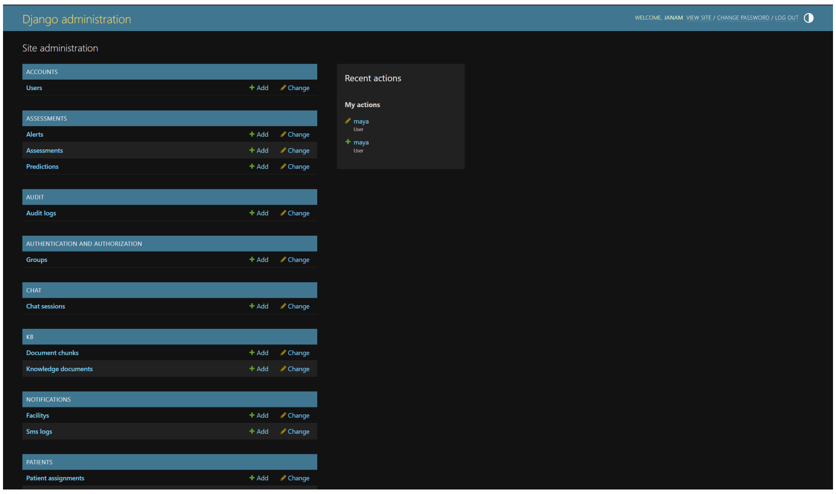


Figure 6.1: Django Administration Panel
The administration panel lists the available data-management modules, including users, assessments, predictions, audit logs, and patient assignments.

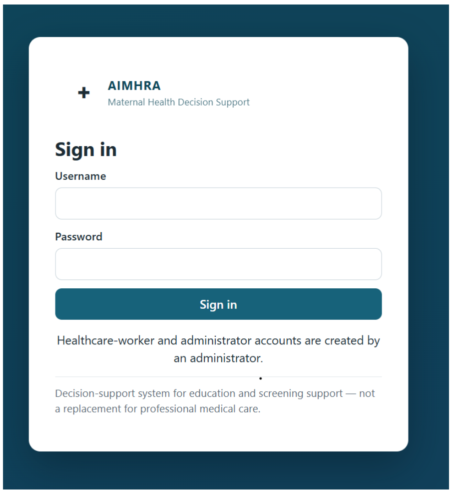


Figure 6.2: System Sign-In Page
The sign-in page provides username and password fields for administrator and healthcare-worker accounts created by an administrator.

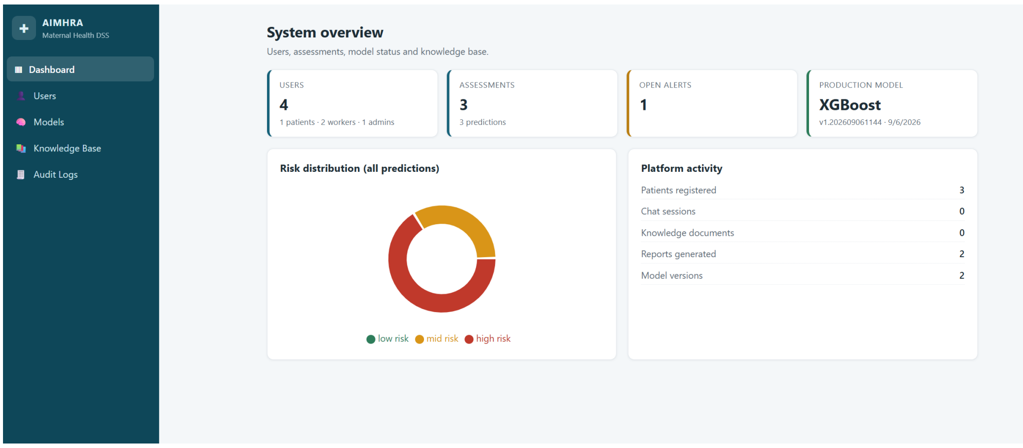


Figure 6.3: Admin Dashboard

The dashboard summarizes users, assessments, open alerts, the production model, prediction risk distribution, and platform activity.

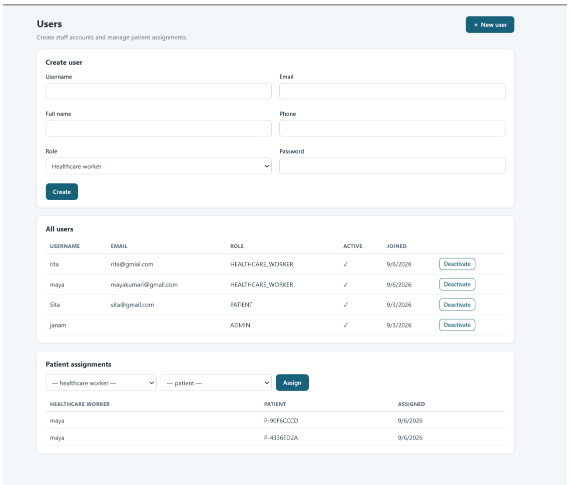


Figure 6.4: User Management and Patient Assignment

This interface provides a user-creation form, account activation controls, and options for assigning patients to healthcare workers.

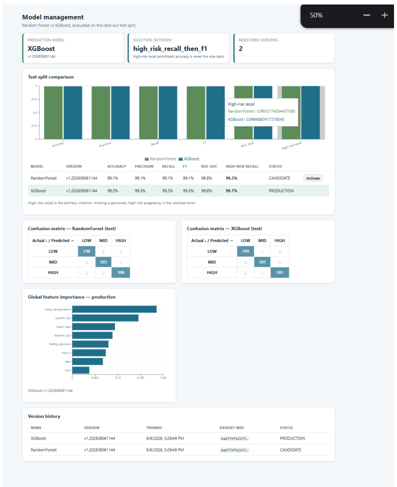


Figure 6.5: Model Management and Performance Comparison
The model-management page displays evaluation metrics, confusion matrices, feature importance, and version history for comparing available models.

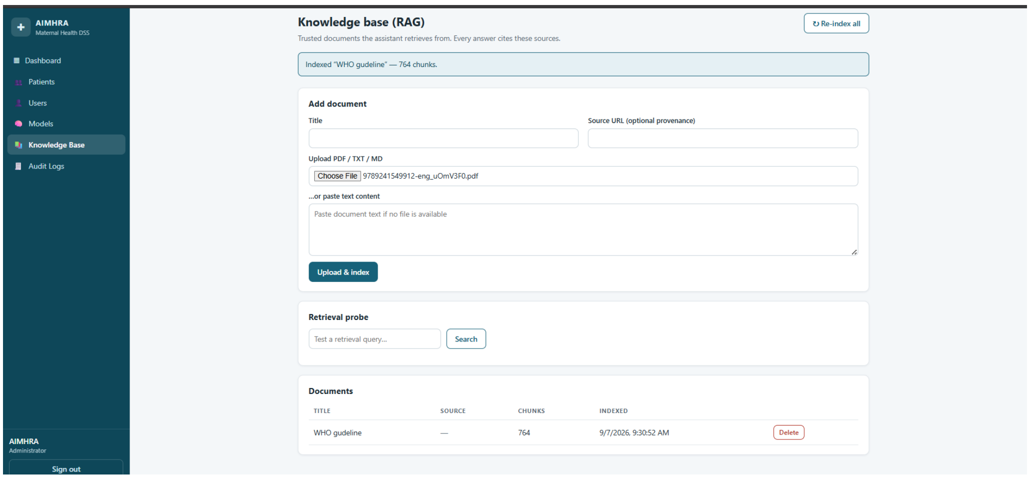
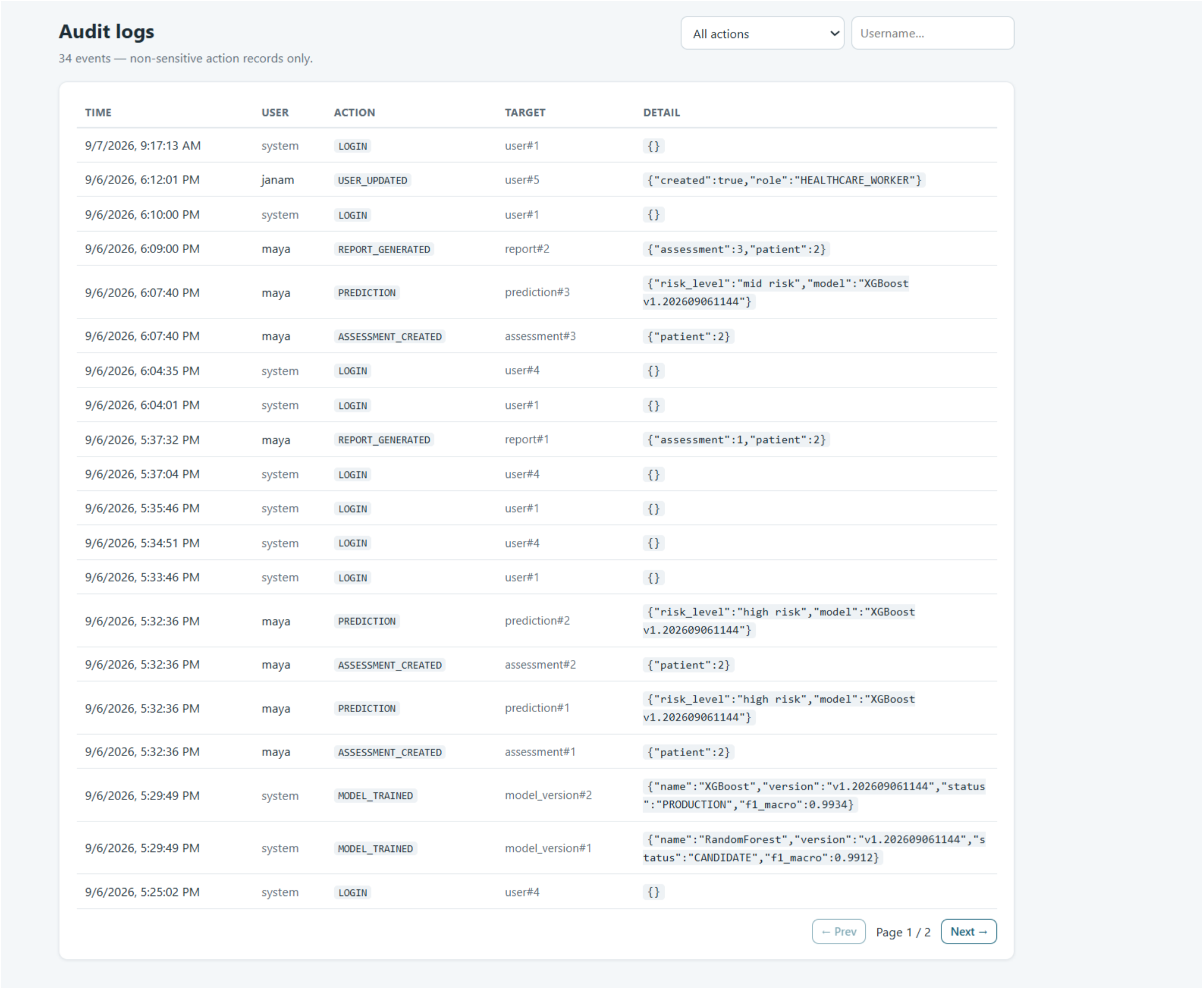


Figure 6.6: Knowledge Base Management
Administrators can upload or enter reference documents, index their contents, and test retrieval queries through the knowledge-base interface.



Audit logs

34 events — non-sensitive action records only.

All actions | Username...

TIME	USER	ACTION	TARGET	DETAIL
9/7/2026, 9:17:13 AM	system	LOGIN	user#1	{}
9/6/2026, 6:12:01 PM	janam	USER_UPDATED	user#5	{"created":true,"role":"HEALTHCARE_WORKER"}
9/6/2026, 6:10:00 PM	system	LOGIN	user#1	{}
9/6/2026, 6:09:00 PM	maya	REPORT_GENERATED	report#2	{"assessment":3,"patient":2}
9/6/2026, 6:07:40 PM	maya	PREDICTION	prediction#3	{"risk_level":"mid risk","model":"XGBoost v1.202609061144"}
9/6/2026, 6:07:40 PM	maya	ASSESSMENT_CREATED	assessment#3	{"patient":2}
9/6/2026, 6:04:35 PM	system	LOGIN	user#4	{}
9/6/2026, 6:04:01 PM	system	LOGIN	user#1	{}
9/6/2026, 5:37:32 PM	maya	REPORT_GENERATED	report#1	{"assessment":1,"patient":2}
9/6/2026, 5:37:04 PM	system	LOGIN	user#4	{}
9/6/2026, 5:35:46 PM	system	LOGIN	user#1	{}
9/6/2026, 5:34:51 PM	system	LOGIN	user#4	{}
9/6/2026, 5:33:46 PM	system	LOGIN	user#1	{}
9/6/2026, 5:32:36 PM	maya	PREDICTION	prediction#2	{"risk_level":"high risk","model":"XGBoost v1.202609061144"}
9/6/2026, 5:32:36 PM	maya	ASSESSMENT_CREATED	assessment#2	{"patient":2}
9/6/2026, 5:32:36 PM	maya	PREDICTION	prediction#1	{"risk_level":"high risk","model":"XGBoost v1.202609061144"}
9/6/2026, 5:32:36 PM	maya	ASSESSMENT_CREATED	assessment#1	{"patient":2}
9/6/2026, 5:29:49 PM	system	MODEL_TRAINED	model_version#2	{"name":"XGBoost","version":"v1.202609061144","status":"PRODUCTION","f1_macro":0.9934}
9/6/2026, 5:29:49 PM	system	MODEL_TRAINED	model_version#1	{"name":"RandomForest","version":"v1.202609061144","status":"CANDIDATE","f1_macro":0.9912}
9/6/2026, 5:25:02 PM	system	LOGIN	user#4	{}

← Prev Page 1 / 2 Next →

Figure 6.7: System Audit Logs

The audit-log page records user actions with timestamps, targets, and details, and provides filters for reviewing system activity.

6.2 Healthcare Worker Interface

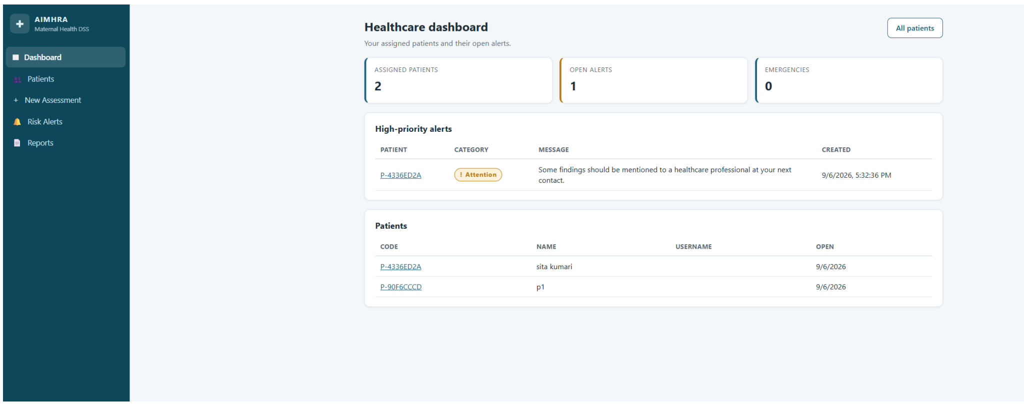


Figure 6.8: Healthcare Worker Dashboard

The dashboard summarizes assigned patients, open alerts, and emergencies, with a high-priority alert list and links to patient records.

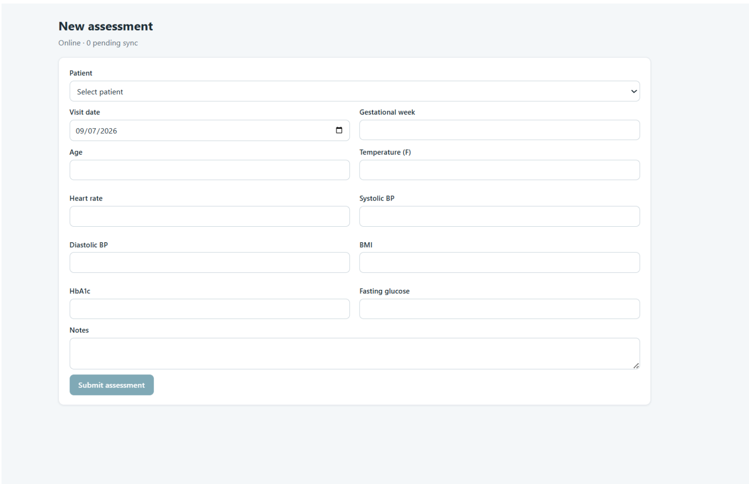


Figure 6.9: New Patient Assessment Form

The assessment form captures the patient, visit date, gestational week, clinical measurements, and notes before submitting an assessment.

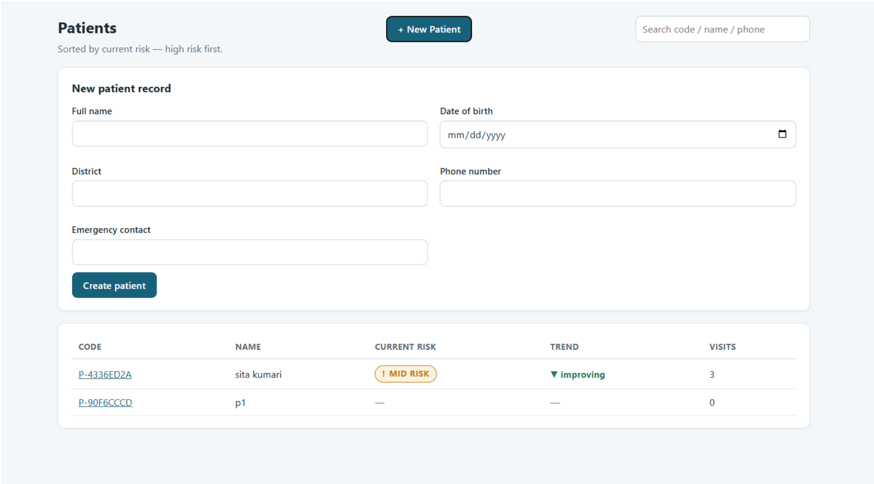


Figure 6.10: Patient Registration

The patient-registration form collects identifying and contact information, including date of birth, district, phone number, and emergency contact.

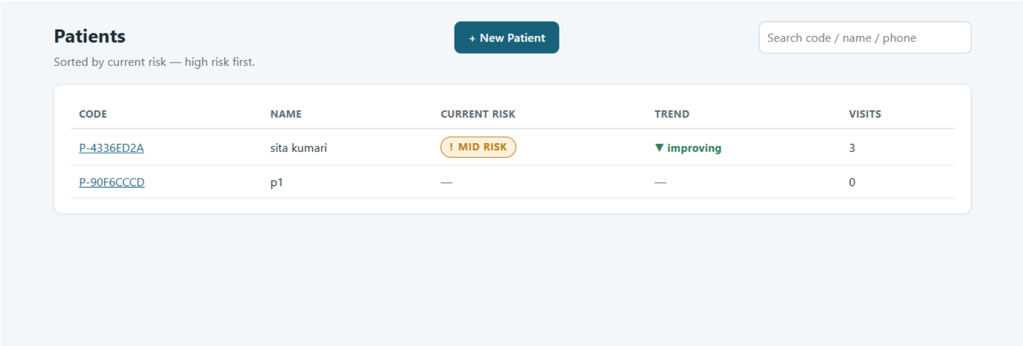


Figure 6.11: Patient List and Risk Trends

The patient list displays each patient's code, name, current risk category, risk trend, and visit count, with a search field for locating records.

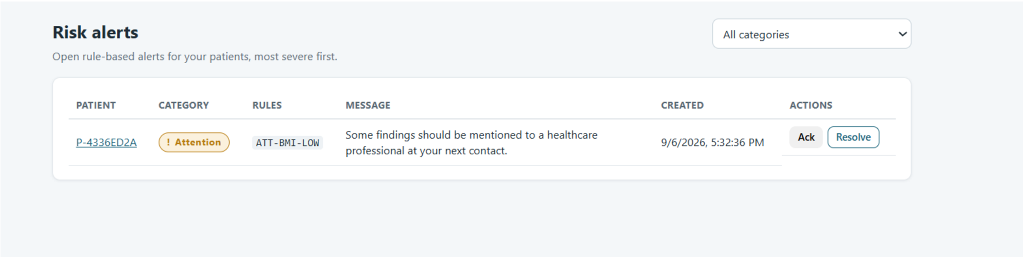


Figure 6.12: Risk Alert Management

The risk-alert page lists patient alerts with their category, triggering rule, message, and creation time, alongside acknowledge and resolve actions.

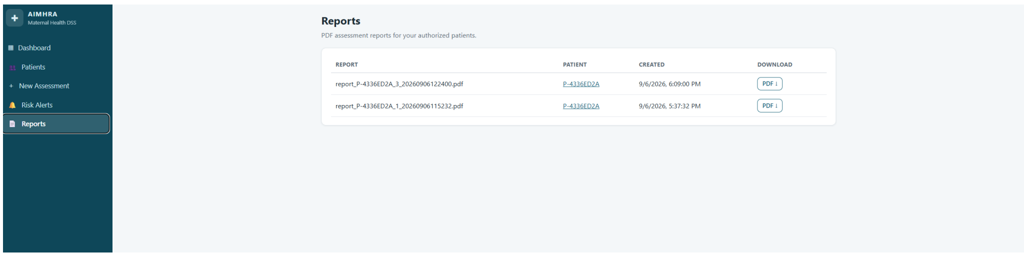
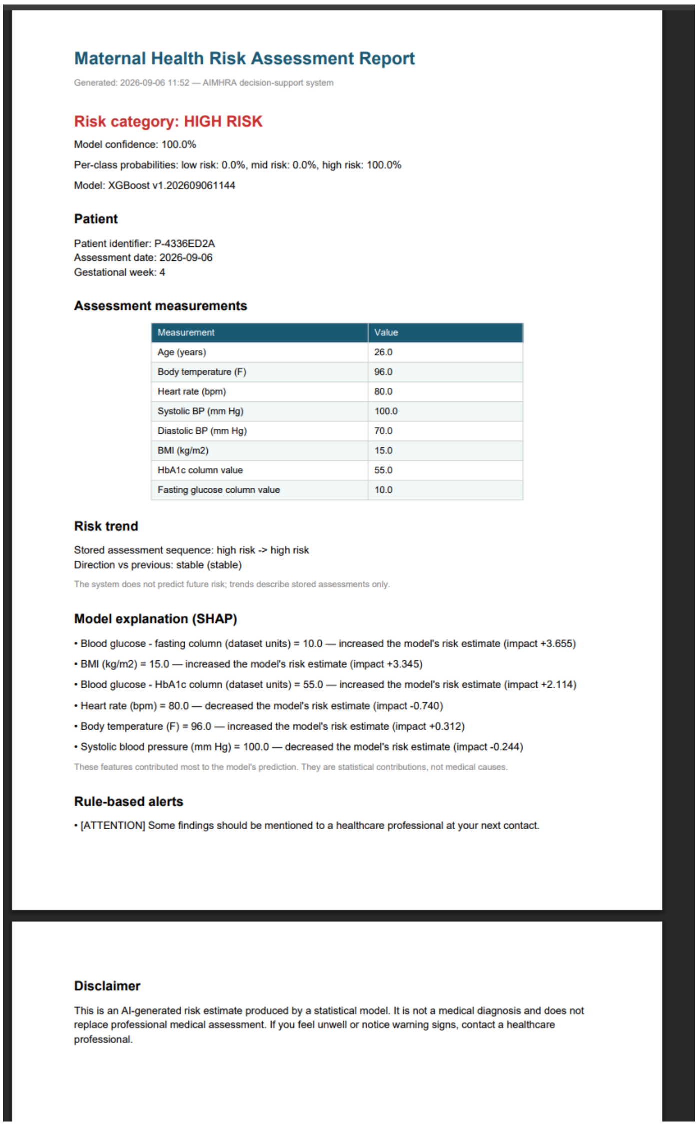


Figure 6.13: Assessment Report Downloads
The reports page lists generated assessment reports for authorized patients and provides PDF download controls.



Maternal Health Risk Assessment Report

Generated: 2026-09-06 11:52 — AIMHRA decision-support system

Risk category: HIGH RISK

Model confidence: 100.0%

Per-class probabilities: low risk: 0.0%, mid risk: 0.0%, high risk: 100.0%

Model: XGBoost v1.202609061144

Patient

Patient identifier: P-4336ED2A
Assessment date: 2026-09-06
Gestational week: 4

Assessment measurements

Measurement	Value
Age (years)	26.0
Body temperature (F)	96.0
Heart rate (bpm)	80.0
Systolic BP (mm Hg)	100.0
Diastolic BP (mm Hg)	70.0
BMI (kg/m2)	15.0
HbA1c column value	55.0
Fasting glucose column value	10.0

Risk trend

Stored assessment sequence: high risk -> high risk
Direction vs previous: stable (stable)

The system does not predict future risk; trends describe stored assessments only.

Model explanation (SHAP)

• Blood glucose - fasting column (dataset units) = 10.0 — increased the model's risk estimate (impact +3.655)
• BMI (kg/m2) = 15.0 — increased the model's risk estimate (impact +3.345)
• Blood glucose - HbA1c column (dataset units) = 55.0 — increased the model's risk estimate (impact +2.114)
• Heart rate (bpm) = 80.0 — decreased the model's risk estimate (impact -0.740)
• Body temperature (F) = 96.0 — increased the model's risk estimate (impact +0.312)
• Systolic blood pressure (mm Hg) = 100.0 — decreased the model's risk estimate (impact -0.244)

These features contributed most to the model's prediction. They are statistical contributions, not medical causes.

Rule-based alerts

• [ATTENTION] Some findings should be mentioned to a healthcare professional at your next contact.

Disclaimer

This is an AI-generated risk estimate produced by a statistical model. It is not a medical diagnosis and does not replace professional medical assessment. If you feel unwell or notice warning signs, contact a healthcare professional.

Figure 6.14: Generated Maternal Health Risk Assessment Report
The generated report presents the risk category, patient information, recorded measurements, risk trend, SHAP-based explanation, advisory text, and a clinical disclaimer.

6.3 Data Set Analysis and Visualization

6.3.1 Distribution of Risk Classes

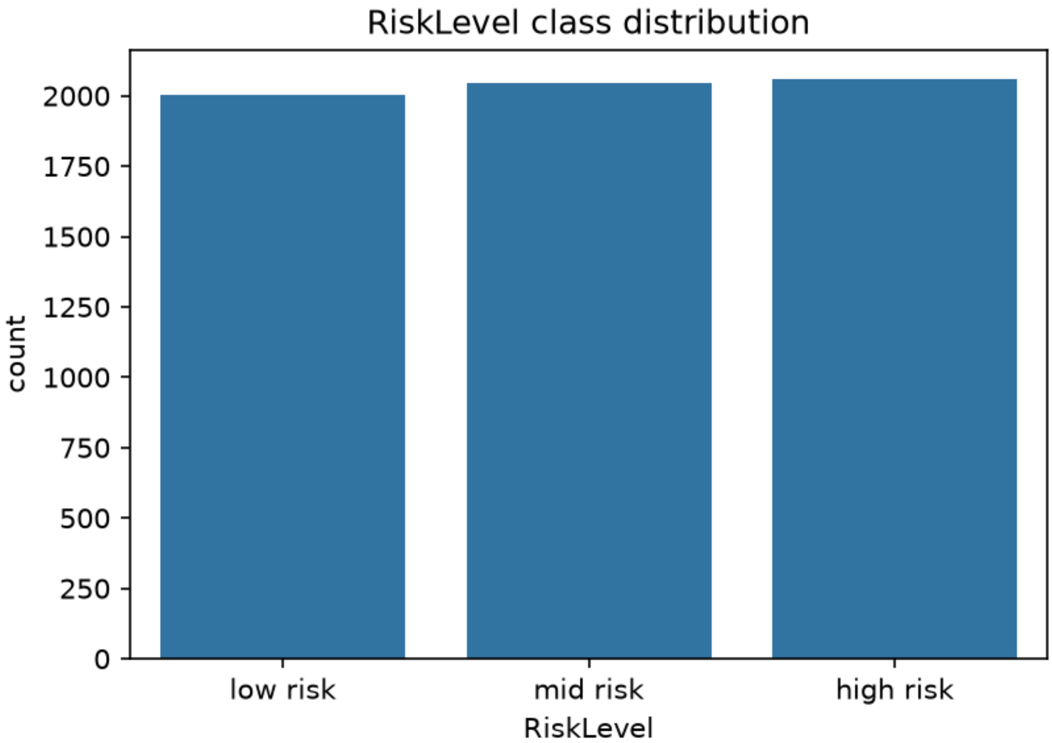


Figure 6.15: Risk class distribution

The bar chart shows the number of records in the low-, mid-, and high-risk categories. Each class contains approximately 2,000 records, with the high-risk class having slightly more records than the mid-risk class and the low-risk class having the fewest. The similar bar heights indicate a nearly balanced class distribution in the data shown, with no single risk category dominating the dataset.

6.3.2 Correlation Heatmap

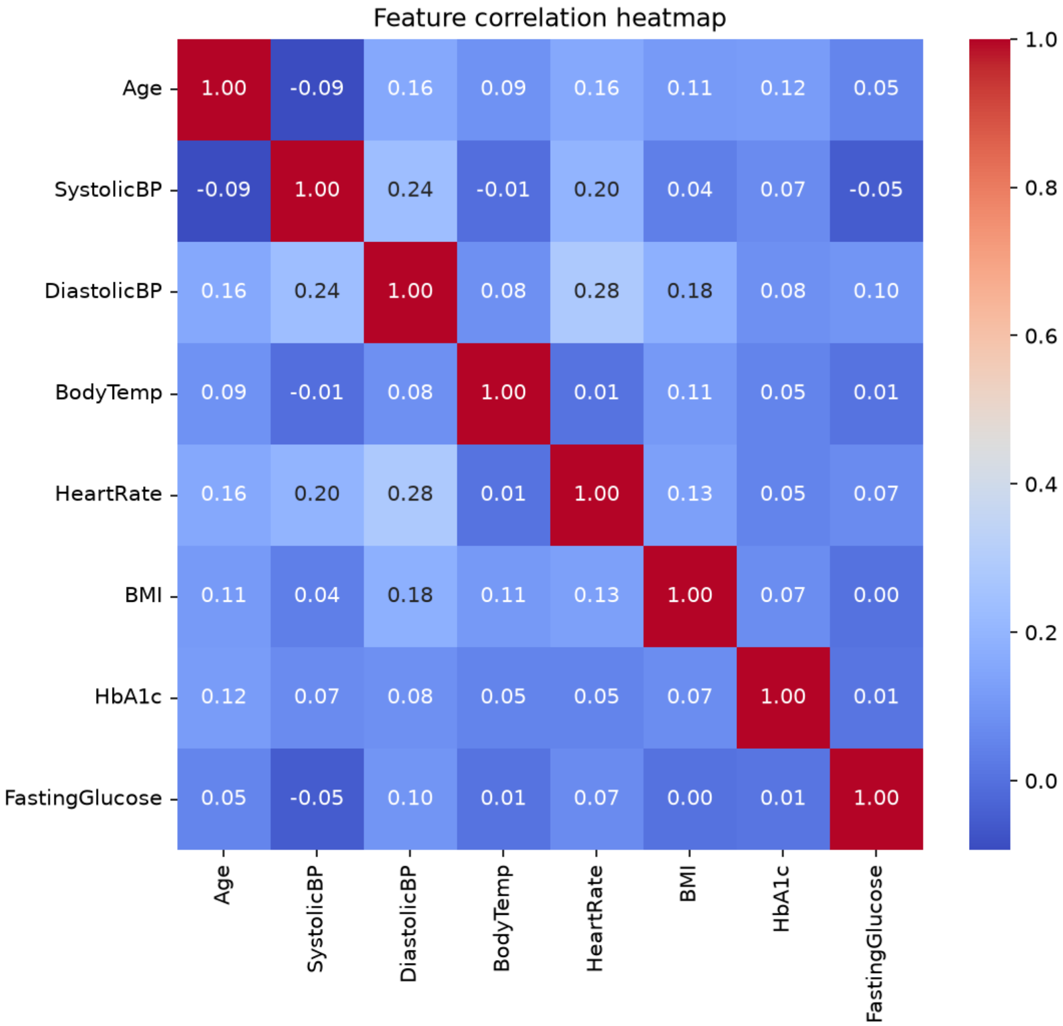


Figure 6.16: Correlation Heatmap

The heatmap summarizes pairwise correlations among age, systolic and diastolic blood pressure, body temperature, heart rate, body mass index (BMI), HbA1c, and fasting glucose. The diagonal values are 1.00 because each feature is perfectly correlated with itself. Correlations between different features range from -0.09 to 0.28, indicating generally weak associations. The strongest positive correlation is between diastolic blood pressure and heart rate (0.28), followed by systolic and diastolic blood pressure (0.24). Age and systolic blood pressure show a weak negative correlation (-0.09), while BMI and fasting glucose have a correlation rounded to 0.00. Overall, no feature pair shows a strong correlation; however, these values alone do not establish feature independence or indicate predictive importance for maternal risk classification.

6.4 Machine Learning Models Results

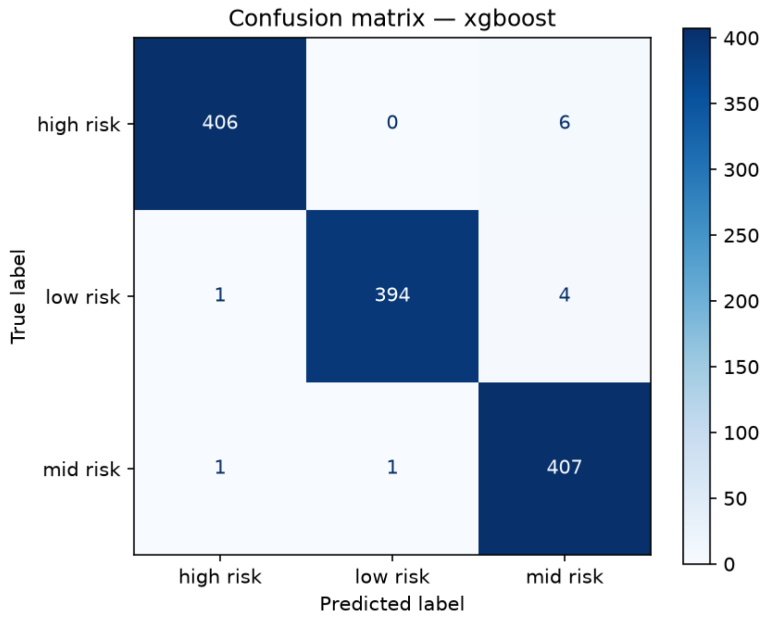


Figure 6.17: Confusion Matrix of the XGBoost Model

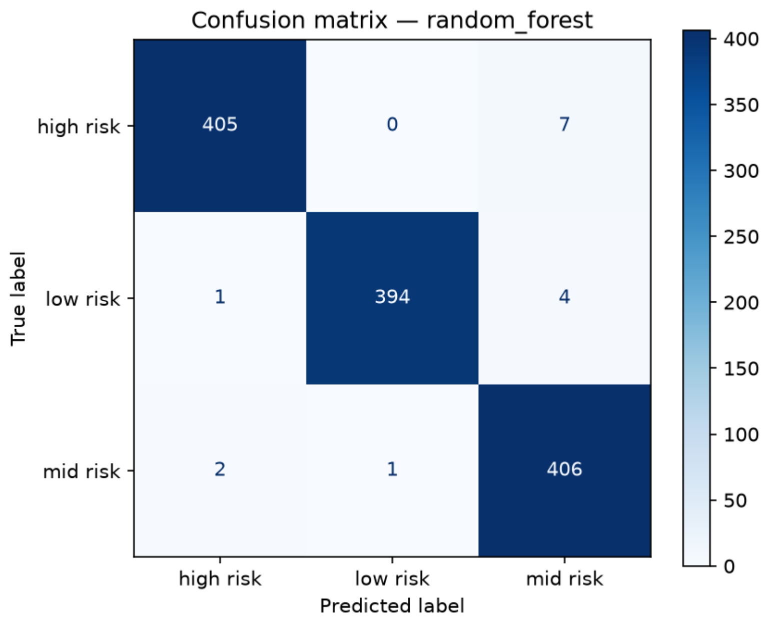


Figure 6.18: Confusion Matrix of the Random Forest Model

6.4.1 Random Forest and XGBoost Performance Comparison

Table 6.1 summarizes the reported test-split metrics for model version `v1.202609070354` of each model, as recorded in `mlcore_modelversion`. Both models were evaluated on the same held-out test split, comprising 15% of the dataset. Overall precision, recall, and F1-score are macro-averaged.

Table 6.1: Random Forest and XGBoost Test-Split Performance

Metric	Random Forest	XGBoost
Model status	Candidate	Production
Accuracy	0.9912 (99.12%)	0.9934 (99.34%)
Precision (macro)	0.9912	0.9934
Recall (macro)	0.9912	0.9934
F1-score (macro)	0.9912	0.9934
High-risk recall	0.9934	0.9967
ROC-AUC (one-vs-rest macro)	≈ 0.999	≈ 0.999
Per-class F1-score		
Low risk	0.9917	0.9917
Mid risk	0.9870	0.9902
High risk	0.9950	0.9983

Evaluation configuration. Both models used a stratified 70%/15%/15% training/validation/test split with random seed 42 and median imputation. Random Forest used 400 trees, `min_samples_leaf=2`, and `class_weight=balanced_subsample`. XGBoost used 400 trees, a learning rate of 0.08, maximum depth of 6, subsampling of 0.9, and column subsampling of 0.8.

Comparison. XGBoost achieved an accuracy advantage of 0.22 percentage points and a high-risk recall advantage of 0.33 percentage points. It also achieved higher macro-averaged precision, recall, and F1-score, as well as higher mid-risk and high-risk F1-scores. Low-risk F1-score was identical, while the reported approximate ROC-AUC values do not distinguish the models. XGBoost was registered as the production model using the criterion `high_risk_recall_then_f1`, which prioritizes detection of high-risk cases before overall F1-score; Random Forest remained a candidate.

6.5 Feature Importance and Model Explainability

6.5.1 Feature Importance Comparison

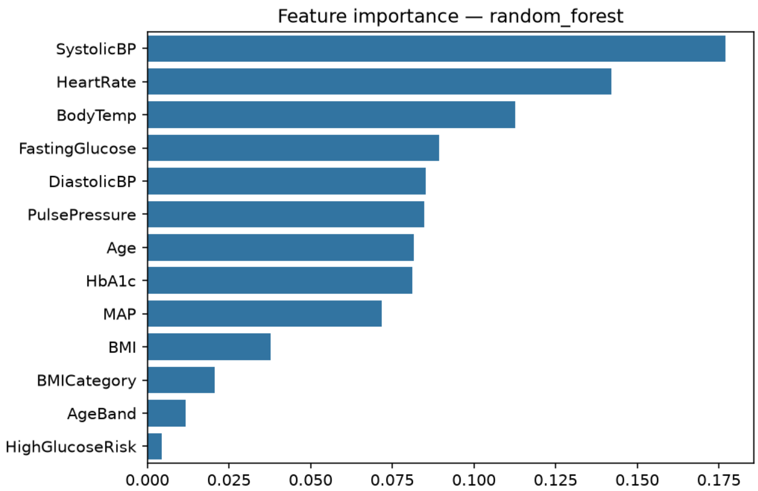


Figure 6.19: Feature Importance of the Random Forest Model

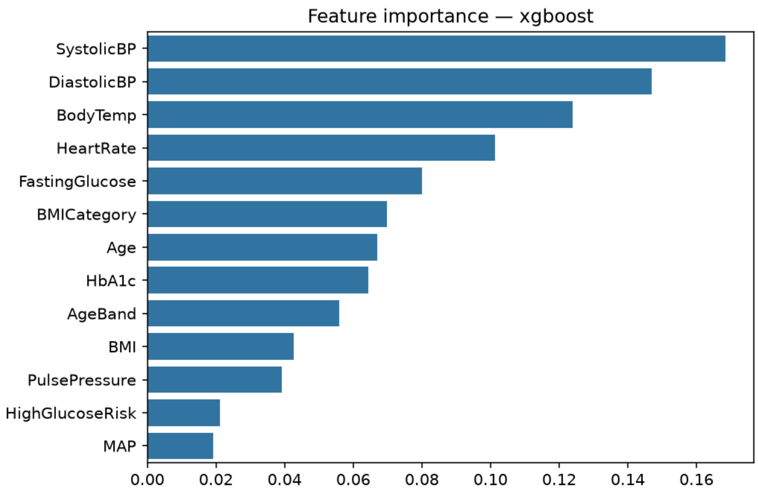


Figure 6.20: Feature Importance of the XGBoost Model

Figures 6.19 and 6.20 both rank systolic blood pressure as the most important feature. Random Forest places heart rate second, followed by body temperature and fasting glucose. XGBoost instead places diastolic blood pressure second, followed by body temperature and heart rate. Pulse pressure and mean arterial pressure (MAP) rank higher in Random Forest, whereas BMI category and age band rank higher in XGBoost. The high-glucose-risk indicator has relatively low importance in both models. These plots summarize the reported feature importance, but do not show whether a feature increases or decreases a particular class output or establish a causal relationship.

6.5.2 SHAP Summary of Random Forest

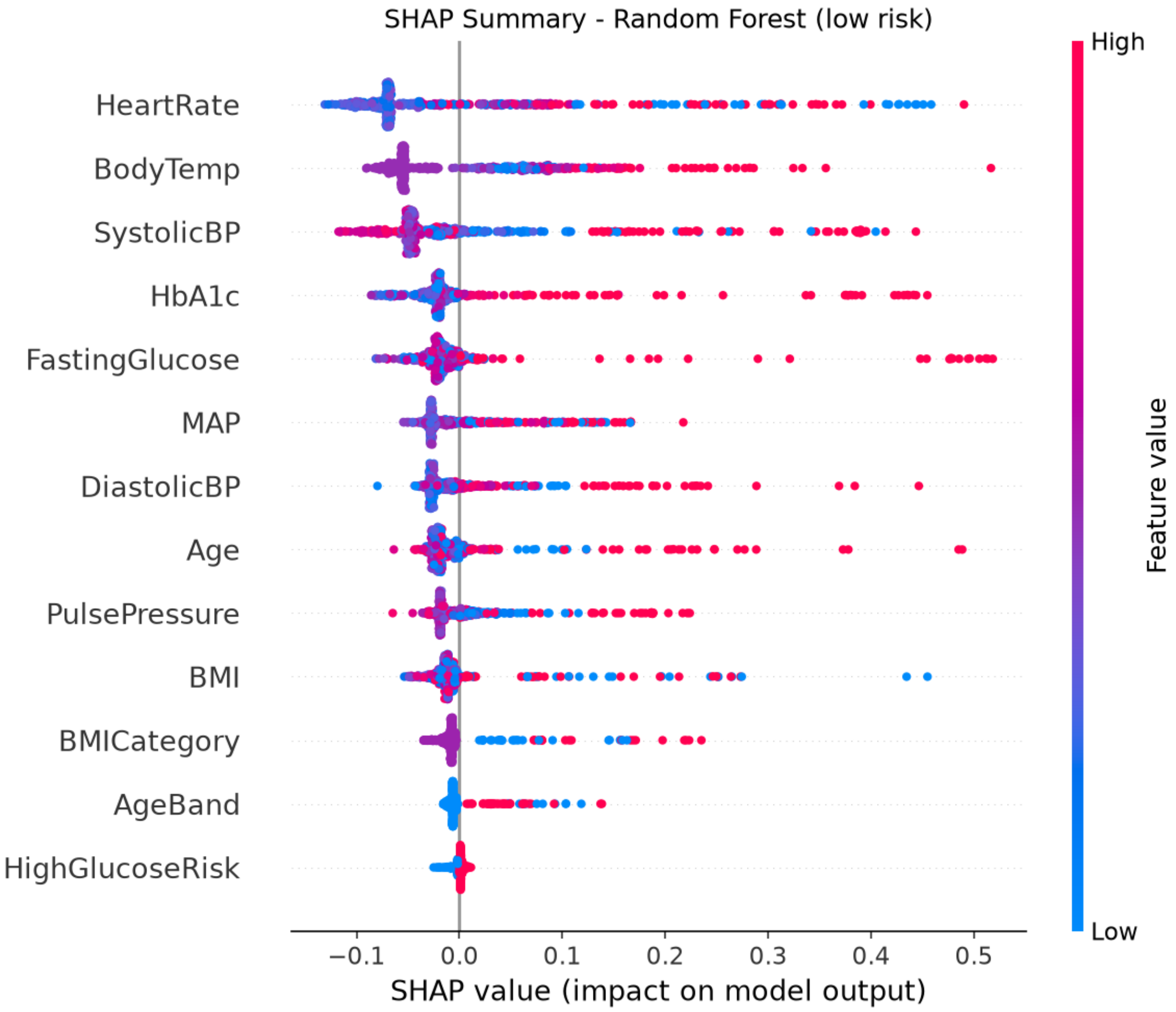


Figure 6.21: Random Forest SHAP Summary for the Low-Risk Class

In the supplied SHAP summaries, the horizontal axis shows the contribution to the class output named in the plot title: positive values support that class output, while negative values oppose it. Red points indicate higher feature values and blue points indicate lower values; the colour does not denote the risk class.

Figure 6.21 places heart rate, body temperature, and systolic blood pressure at the top of the low-risk summary. These features show substantial horizontal spread, while HbA1c and fasting glucose also have pronounced positive tails for some observations. Age band and the high-glucose-risk indicator have comparatively small contributions. The mixed colours across several rows do not support a simple rule that a higher feature value always increases or decreases the low-risk output.

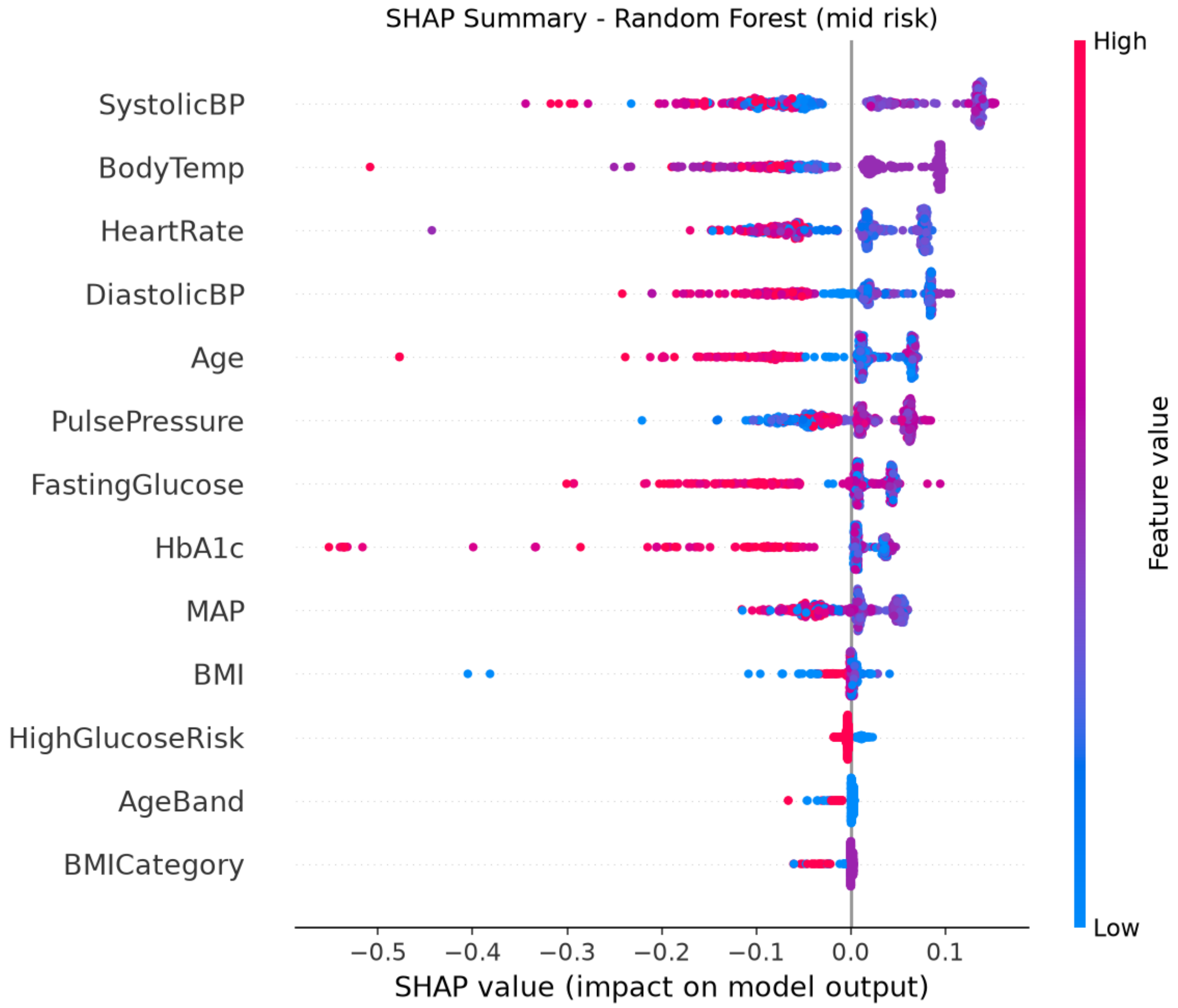


Figure 6.22: Random Forest SHAP Summary for the Mid-Risk Class

Figure 6.22 places systolic blood pressure, body temperature, and heart rate first in the mid-risk summary, followed by diastolic blood pressure and age. Several features have longer negative than positive tails, indicating that some observations receive substantial contributions away from the mid-risk output. Higher values of fasting glucose and HbA1c are prominent in their negative tails, although many observations remain near zero. BMI category, age band, and the high-glucose-risk indicator have relatively small contributions in this plot.

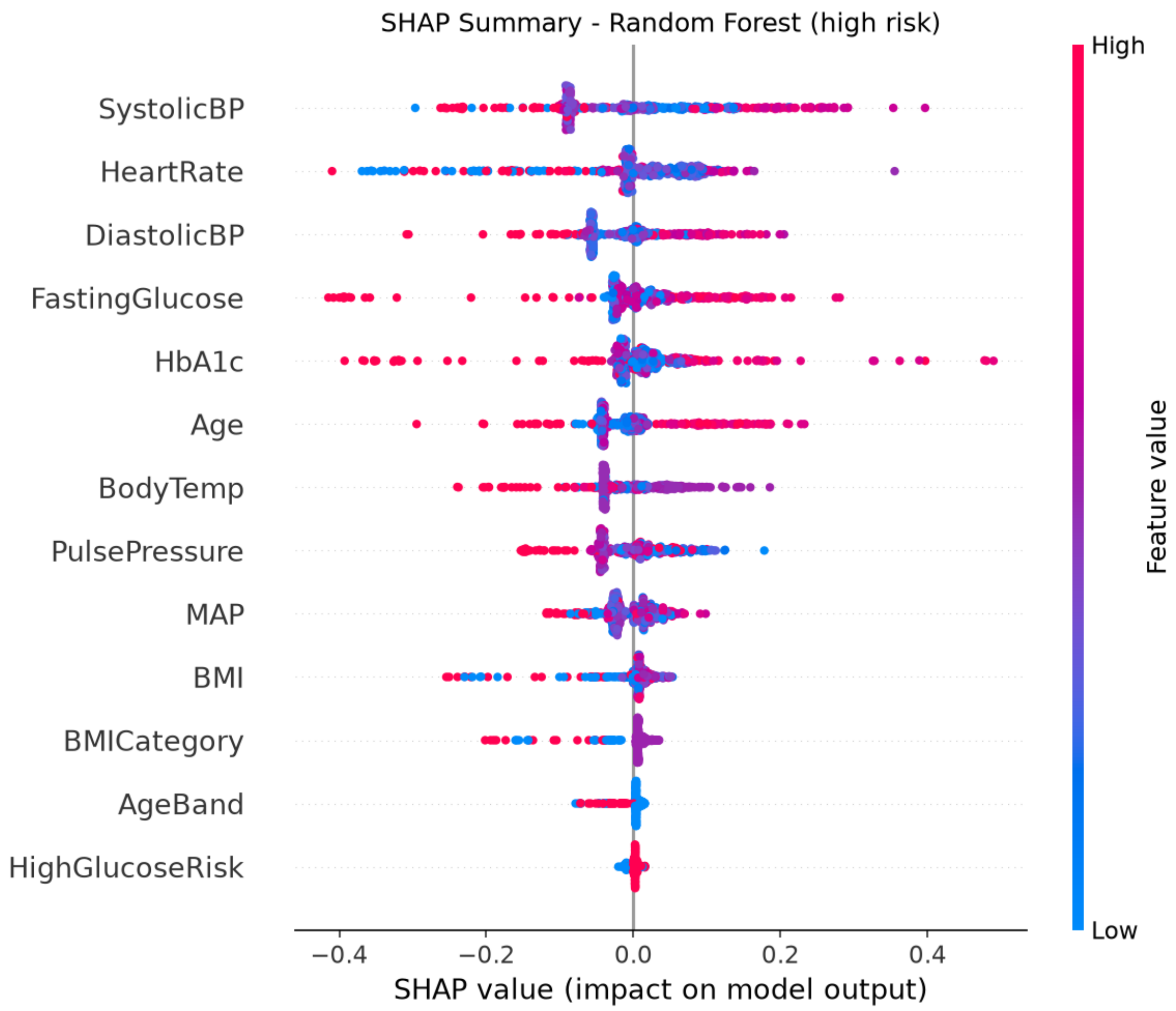


Figure 6.23: Random Forest SHAP Summary for the High-Risk Class

Figure 6.23 places systolic blood pressure and heart rate at the top of the high-risk summary, followed by diastolic blood pressure, fasting glucose, and HbA1c. Systolic blood pressure and the glucose-related measurements show contributions on both sides of zero, with some large positive contributions for the high-risk output. Heart rate also shows a broad negative tail. BMI category, age band, and the high-glucose-risk indicator are more concentrated around zero. The presence of high-valued points on both sides of zero for several features means that their displayed effects are not uniformly directional.

6.5.3 SHAP Summary of XGBoost

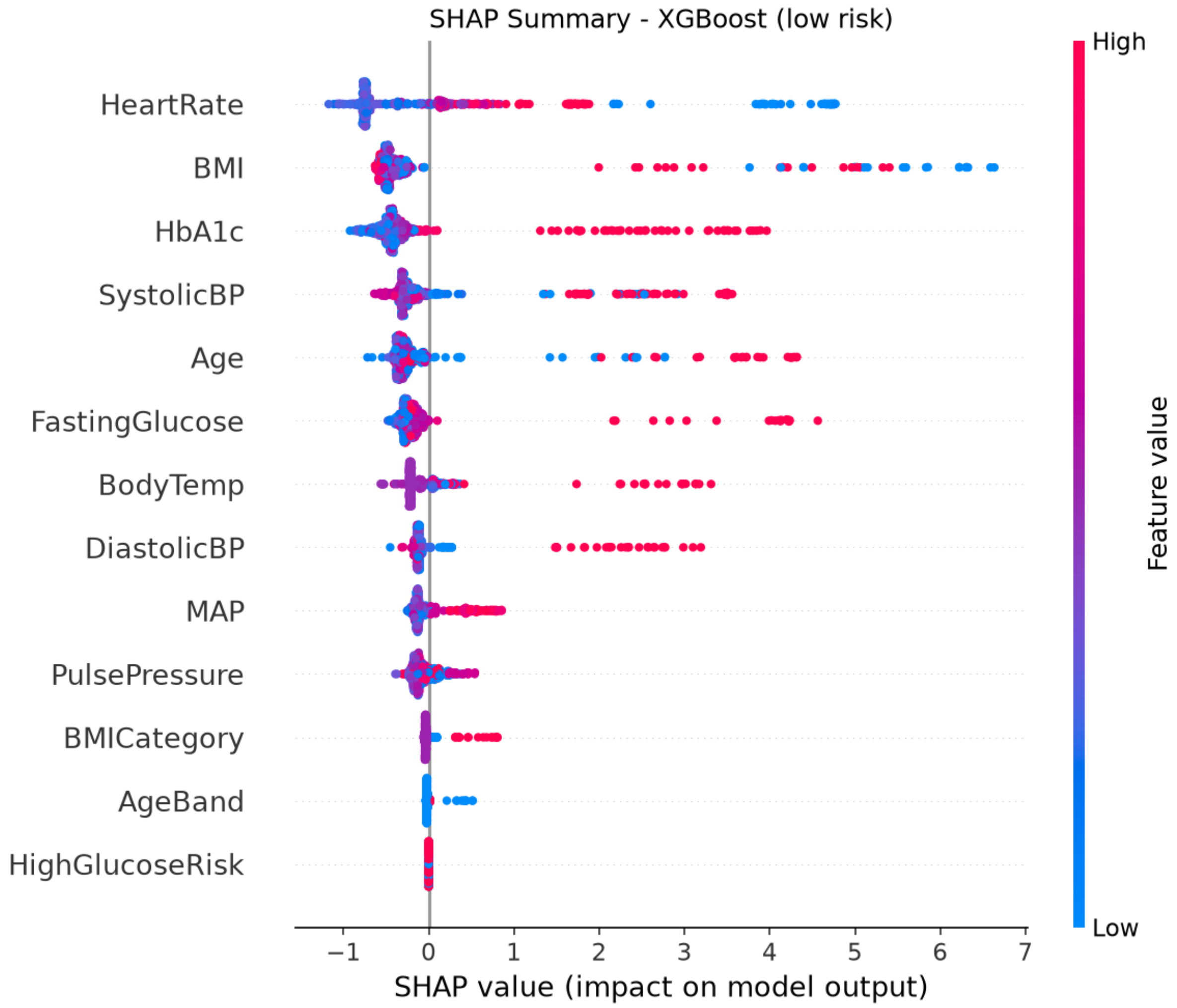


Figure 6.24: XGBoost SHAP Summary for the Low-Risk Class

Figure 6.24 places heart rate, BMI, and HbA1c at the top of the low-risk summary, followed by systolic blood pressure and age. BMI has a particularly long positive tail, while HbA1c, fasting glucose, and several other features also show large positive contributions for subsets of observations. Many points remain clustered near zero or slightly below it. BMI category, age band, and the high-glucose-risk indicator have comparatively limited spread. Notably, BMI appears much higher in this class-specific summary than in the overall XGBoost feature-importance plot.

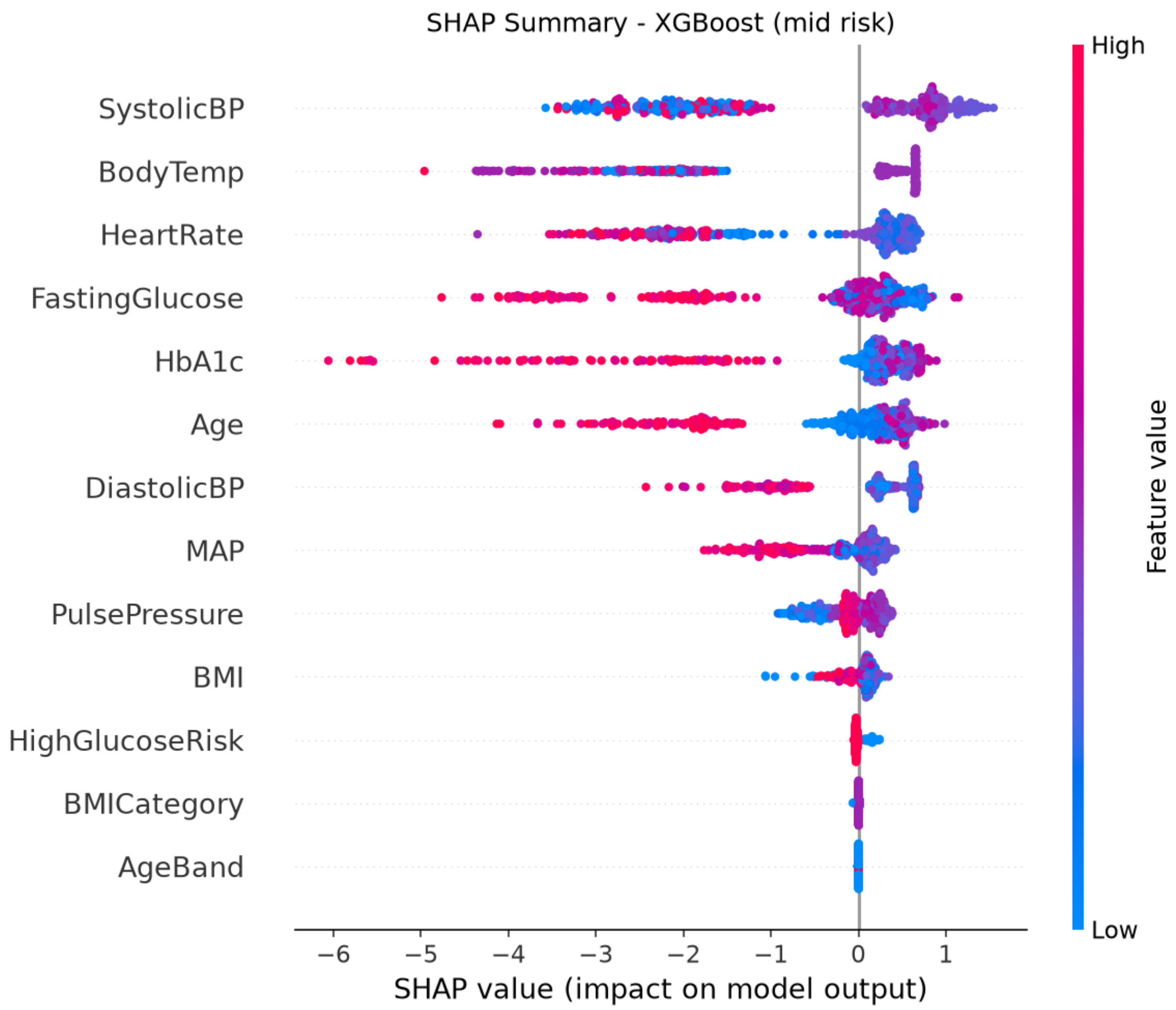


Figure 6.25: XGBoost SHAP Summary for the Mid-Risk Class

Figure 6.25 places systolic blood pressure, body temperature, and heart rate first, matching the leading three features in the Random Forest mid-risk summary. Fasting glucose and HbA1c follow. Several rows show substantial negative contributions, with higher HbA1c, fasting glucose, and age values prominent in the negative tails. Positive contributions are generally more compact. BMI category and age band are almost entirely concentrated at zero, and the high-glucose-risk indicator has only a small spread in the displayed output scale.

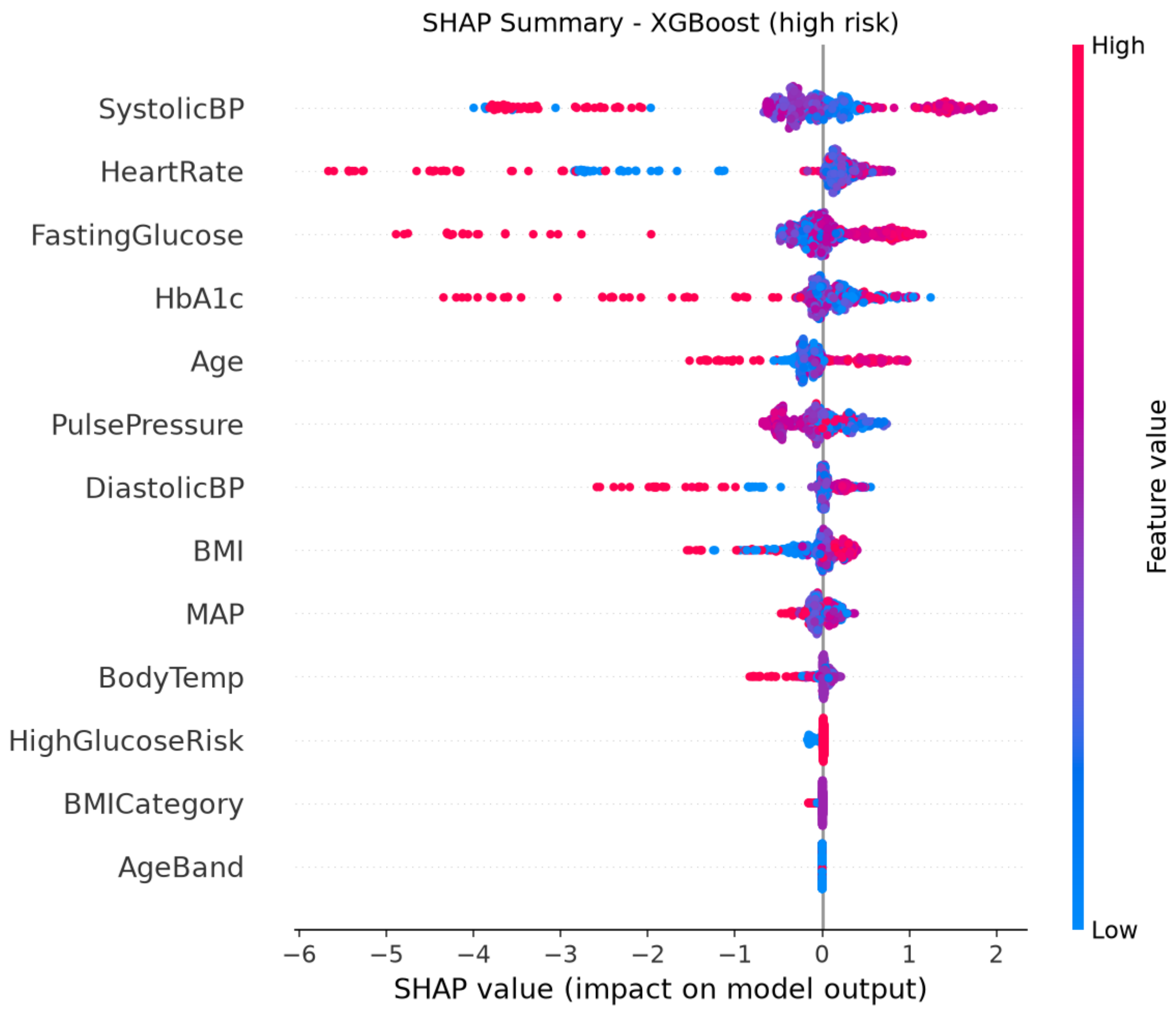


Figure 6.26: XGBoost SHAP Summary for the High-Risk Class

Figure 6.26 places systolic blood pressure and heart rate first, followed by fasting glucose, HbA1c, and age. Systolic blood pressure has a clear positive cluster as well as a long negative tail. Heart rate, fasting glucose, and HbA1c also show substantial negative contributions for some observations, alongside points supporting the high-risk output. Body temperature has a lower position here than in the overall feature-importance ranking. The engineered indicators near the bottom have little visible spread.

The XGBoost plots use a much wider numerical SHAP range than the Random Forest plots. Without confirmation of a common explained-output scale, these magnitudes should not be compared directly across models. The descriptions follow the class labels in the supplied images and characterize model behaviour rather than clinical or causal effects.

Chapter 7

Work to Be Done

- **AI Chat Assistant**: Extend the current RAG-based assistant to incorporate patient-specific context, multi-turn conversation memory, and integration with the Gemini API as originally scoped.
- **Offline inference on the client device**: The system currently supports offline-first data capture and sync only; true on-device ML inference via TensorFlow.js or ONNX was not implemented due to model conversion trade-offs, and remains a candidate for future work.
- **SMS alerting — production credentials**: The notification module is built and tested against the Sparrow SMS API in sandbox mode; full verification with live credentials and real facility contacts is recommended before field deployment.
- **Model retraining pipeline**: Introduce an automated retraining workflow that incorporates newly collected field data to periodically update and validate the deployed model version.
- **Formal usability testing with FCHVs**:Conduct field-based usability testing with actual Female Community Health Volunteers to validate the interface for users with varying levels of digital literacy, prior to production deployment.
- **UI improvement**:Simplify the interface with bigger buttons, clearer layout, and easier navigation so it is simple to use for FCHVs with basic smartphone experience.

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